(In)sensitivity to surface-level heuristics: A case from Turkish verbal attractors

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Abstract

The linguistic illusions literature has debated what information is used to access memory representations. Structural, semantic, and discourse cues are known to guide the formation of subject-verb agreement dependencies, but it remains unclear how native speakers use surface level phonological information during dependency resolution. Agreement attraction is more likely when the attractor has the same form as an agreement controller would (e.g., in case-marking). Slioussar (2018) attributes this to a phonological form directly activating all feature sets that it can express, independent of context. An alternative proposal is that phonological form impacts agreement indirectly, by increasing or decreasing the probability that a given noun could be a subject, or an agreement controller. We investigate these possibilities in Turkish, where verbal and nominal elements both can surface as subjects and both are marked for number by the same phonological form *-lAr*, but only nominal plural *-lAr* can ever control verbal agreement. Two speeded acceptability studies and a self-paced reading study compared plural-marked genitive noun attractors, which can be subjects and agreement controllers in other environments, with plural-marked verbal attractors, which can be subjects but never agreement controllers. Attraction was robust from genitive nouns and absent from verbal attractors. This pattern is inconsistent with the hypothesis that phonological forms invariantly activate associated feature sets, and suggests that phonological form influences agreement by affecting the estimated likelihood of the attractor being a controller, not a subject. The account also explains previously observed discrepancies between form effects on attraction in Russian vs. Czech.

Keywords: form-sensitivity; memory; agreement attraction; linguistic illusions; sentence processing; Turkish

1 Introduction

A substantial body of work has shown that sentence comprehension and production are sensitive not only to syntactic and semantic properties of the input, but also to its surface form—the phonological or orthographic shape of the words themselves. Acheson and MacDonald (2011), for example, found that participants showed slower reading times in multi-clause sentences and lower accuracy on comprehension questions when the subjects of the two clauses shared phonological similarity (*baker-banker* vs. *runner-banker*). Related work in short-term memory and word recognition shows similar effects—items that overlap phonologically or morphologically are more confusable and more easily retrieved (Copeland and Radvansky, 2001; Rastle and Davis, 2008). However, it is unresolved whether this heuristic penetrates dependency resolution itself–including subject-verb agreement, pronoun resolution, or the licensing of negative polarity items–beyond general effects on reading ease and memory. For example, Kush et al. (2015) show that participants reading a sentence containing a filler-gap dependency did not show any extra difficulty in retrieval or integration at the dependency site when they were asked to simultaneously hold in memory a list of phonologically-similar words to the filler.

In the current study we explore how exactly surface-level form similarities contribute to the processing of syntactic dependencies, focusing on the case of verbal agreement. Errors of agreement processing called *agreement attraction* occur when participants err in establishing number agreement between a verb and its agreement controller because another noun with a different number (the attractor) interferes (Bock and Miller, 1991; Jäger et al., 2017; Smith and Vasishth, 2020; Phillips et al., 2011). Prior work on agreement attraction has suggested that form similarity of the attractor to agreement controllers in the language modulates error rates (Hartsuiker et al., 2003; Lago et al., 2019), although such effects appear variable across paradigms and languages (Bock and Eberhard, 1993; Lacina and Chromỳ, 2022). However, it is unresolved whether these effects are due to direct, context-independent activation of syntactic feature sets by phonological form, or an indirect influence of phonological form on the estimation of structural roles in the sentence. Here we use evidence from Turkish to argue that phonological form exerts an indirect influence on verbal agreement computation, in affecting the estimated likelihood of the attractor being a controller.

1.1 Agreement attraction

Agreement attraction occurs when speakers produce sentences like (1a) or misclassify them as acceptable compared to the sentences in (1b) in which there is no plural attractor (Bock and Miller, 1991; Pearlmutter et al., 1999). In reading studies, participants typically read sentences like (1a) faster after the number-marking element *are* compared (1b).

(1) a. * The player on the courts are tired from a long-game.

b. * The player on the court are tired from a long-game.

One proposed explanation for agreement attraction is feature percolation. In feature percolation accounts, the plural feature of the attractor is copied and 'passed up' through syntactic projections such that the encoding of the subject's number is corrupted (Bock and Miller, 1991; Bock and Cutting, 1992; Bock and Eberhard, 1993). For example in a phrase such as *the key to the cabinets*, the plural feature of *cabinets* percolates up to *key*, erroneously licensing plural agreement on the verb. This early account was refined into the Marking and Morphing model (Eberhard, 1999; Eberhard et al., 2005), which no longer requires the attractor to be syntactically contained within the subject phrase; instead, the strength of percolation is a graded function of the syntactic distance between the attractor and the subject head. In this model, attraction proceeds in two stages. At the *marking* stage, participants form a conceptual representation of the subject phrase; notional plurality—for instance, the distributive reading available with *The gang on the motorcycles*—can mark the phrase as number-relevant even without a syntactic percolation route. At the *morphing* stage, the number representation of the phrase is reconciled with other sources of number information, weighted by syntactic distance: the further the attractor is from the subject head, the weaker its influence. If the reconciled value is not definitively singular, in other words there is a percolation from a plural source, the verb may be produced or accepted with plural agreement. The Marking and Morphing model captures fundamental patterns of attraction, including the well-known singular/plural asymmetry—plural attractors, being marked, can spread and shift the reconciled value, whereas singular attractors are underspecified and exert minimal influence (Eberhard et al., 2005). While the account was initially developed with production studies in mind, Hammerly et al. (2019) have subsequently argued that many attraction effects in comprehension can also be modeled as a product of representational distortion paired with a priori response biases.

Whereas feature percolation accounts locate the source of attraction in the representation of the subject phrase itself—the inferred number of the subject is distorted—an alternative family of accounts holds that this representation remains intact, and that errors instead arise when the process of accessing it goes awry. In the cue-based retrieval account, the parser stores each noun phrase as a chunk in content-addressable memory, encoding features such as number, case, gender, and syntactic role (Lewis and Vasishth, 2005). When a verb is encountered in an agreement configuration, retrieval cues trigger a search through memory for a chunk matching the profile of an agreement controller. For instance, in the case of English subject-verb agreement within sentences like (1a), the verb provides cues like [+NOM, +PL, +CONTROLLER], and any chunks partially matching these cues are activated and therefore can interfere with retrieval. The retrieval latency at the site of number-bearing element *are* (measured with reading time) and the successful retrieval of the correct controller (measured with grammaticality judgment) are thus contingent on the activation level for each chunk (Lewis and Vasishth, 2005). The activation level of each chunk is calculated according to how many of their features match the retrieval cues and how many other chunks have similar features. In (1a), both the noun *player* and the *courts* partially match with the retrieval cues. The chunk corresponding to *player* matches with the features +NOM and +CONTROLLER, while the *courts* chunk matches with +NOM and +PL. Since there is not a single perfect match, and both NPs have partial matches, in a subset of trials the chunk for *courts* is retrieved as the controller and comprehenders mistakenly deem ungrammatical sentences grammatical. This erroneous retrieval occurs more often in (1a) compared to (1b), because *court* in (1b) matches with fewer cues and less likely to be retrieved. As for reading times, the number-bearing element *are* has been observed to be read faster in (1a) compared to (1b) even though both sentences are ungrammatical (Lewis and Vasishth, 2005; Wagers et al., 2009).

Our aim is not to adjudicate between representational and retrieval-based accounts of attraction in general. On either view, the mechanism must specify what information it consults—the features that enter into marking and morphing, or the cues that guide retrieval—and it is this question our experiments target: whether the surface form of an attractor contributes to attraction directly, or only insofar as that form is associated with being an agreement controller. For concreteness we adopt the vocabulary of cue-based retrieval in what follows, and we return to how each class of accounts fares against our results in the General Discussion.

1.2 Case syncretism and phonological overlap in agreement processes

Investigating the factors that modulate agreement attraction errors has been the focus of much research in psycholinguistics, because error patterns can give insight into the algorithms that support parsing of syntactic dependencies in general. Previous work suggests that phonological/orthographic form is one factor that impacts the frequency of agreement errors. However, despite much relevant research, it remains unclear exactly how native speakers use surface-level form information in agreement computation.

Previous studies have shown that agreement attraction errors are more likely when the attractor has the same form as an agreement controller would in that language. In English, most noun phrases would have the same form whether or not they are in a syntactic position that controls verbal agreement. For example, in 'the girl with the dogs is sleeping', it is the head noun phrase 'the girl' that controls agreement, but 'the dog' would have had the same phonological and orthographic form if it were not the agreement controller (e.g. 'the dogs with the girl are sleeping'). This is called case syncretism, and most noun phrases in English – such as *the cabinet* – are syncretic between nominative and accusative case.[1] In other languages, nouns in different positions may be overtly case-marked with different forms. In Hungarian, for example, the counterpart of our English example is *a lány a kutyákkal alszik* 'the girl with the dogs is sleeping', where the attractor *a kutyákkal* bears the overt instrumental suffix *-kal* and is thus formally distinct from its nominative form *a kutyák* that would appear in subject position, as in *a kutyák a lánnyal alszanak* 'the dogs with the girl are sleeping' (Rounds, 2009).

An early agreement production study in German by Hartsuiker et al. (2003) tested the hypothesis that agreement attraction errors would be more common when the attractor noun phrase had syncretic case–that is, when the attractor noun phrase had the same form as it would have had if it were in an agreement controller position. In German, accusative and nominative plural determiner forms are identical (syncretic) in the feminine, but the dative determiner has a different form. In the other genders (masculine and neuter), accusative and nominative plural determiners are not syncretic. They found that participants produced more agreement errors when the preambles contained two noun phrases whose determiners were not distinctively marked, as in (2a), compared to cases where the attractor could be distinguished by form alone, as in (2b). This additive effect was limited to feminine nouns, the only gender showing nominative–accusative syncretism in plural forms while other nouns showed the base effect of plural.

(2) a. Die Stellungnahme gegen die Demonstration-en
the.F.NOM.SG position against the.F.ACC.PL demonstration-PL
'The position against the demonstrations'

[1]Exceptions are some pronouns which differ in their nominative and accusative forms, as with the first person pronoun *I* (nominative) vs. *me* (accusative).

b. Die Stellungnahme zu den Demonstration-en
the.F.NOM.SG position on the.F.DAT.PL demonstration-PL
'The position on the demonstrations'

Parallel results are reported in a number of comprehension studies. For example, Chromý et al. (2023) conducted four self-paced reading experiments in Czech on agreement attraction configurations, where attraction was measured as faster reading times at the ungrammatical plural verb when the attractor was plural vs. singular. They found attraction effects only when the accusative attractor noun was case syncretic with the nominative agreement controller as in (3a), but not when the attractor had an unambiguous accusative form (3b). Lacina et al. (2025) found similar effects in gender agreement comprehension in Czech in a study based on earlier Slovak production findings (Badecker and Kuminiak, 2007). They found a clear gender attraction effect, as indexed by faster reading times at the ungrammatical verb with the wrong gender agreement when the verb's gender matched the attractor compared to when it did not. However, this attraction effect was only present in cases where the attractor was case syncretic with the nominative agreement controller, but not when the attractor was unambiguously marked with a different case. Slioussar (2018) found similar effects of syncretism on Russian agreement attraction across production, self-paced reading, and acceptability judgments. She compared sentences with accusative plural attractors, which are syncretic with nominative plural forms, to sentences with unambiguously-marked genitive plural attractors. She found that sentences with syncretic accusative plural attractors yielded more plural-marked completions in production, faster reading times at the ungrammatical plural-marked verb in comprehension and higher rates of acceptability judgment errors compared to sentences with unambiguous genitive plural attractors.

(3) a. * Složk-a pro archivářk-y nejspíš bud-ou zahrnovat veškeré nálezy
file-NOM.SG from archiver-ACC.PL$_{(=\text{NOM.PL})}$ probably will-PL include all findings
'A file for archivers$_{\text{ACC}}$ will$_{\text{PL}}$ probably include all findings.'

b. * Pohled od kamarád-ů určitě bud-ou probouzet kásné vzpomínky
postcard[NOM.SG] from friend-GEN.PL$_{(\neq \text{NOM.PL})}$ surely will-PL arouse nice memories
'A postcard from friends$_{\text{GEN}}$ surely will$_{\text{PL}}$ arouse nice memories.'

However, studies in other case-marking languages have demonstrated different patterns of effects for case syncretism in agreement attraction. Franck et al. (2010) investigated agreement production in French with preamble competion, comparing preambles containing case-ambiguous NPs (*le professeur de l'élève/des élèves* 'the professor of the student/students') vs. preambles containing unambiguously accusative-marked pronominal attractors (*le professeur le/les* 'the professor it/them')). They showed that unambiguous marking, i.e. non-syncretic cases, had an increased rate of attraction effects, contrary to the previous results. An agreement comprehension study in Armenian by Avetisyan et al. (2020) found that modulating case ambiguity on the attractor in Armenian had no effect on reading times or error rates. This mixed pattern of case syncretism effects might suggest that these effects are not reducible to surface form ambiguity alone.

A second kind of investigation of the impact of surface form on agreement attraction has tested cases of form overlap between singular and plural, rather than between controller and non-controller forms. In the studies on case syncretism discussed above, the attractor's case may be ambiguous, but its form is not. For example, English *cabinets* is syncretic between nominative and accusative but is not syncretic in form with the singular *cabinet* (in contrast to rare but existing number syncretic forms like *sheep*). In addition to the case-syncretism contrast reviewed above, Slioussar's study Slioussar (2018) included an elegant comparison of a Russian syncretism pattern that crosses case and number. In Russian, not only are many noun forms ambiguous between nominative and genitive case, but for some paradigms the case ambiguity goes together with number ambiguity, yielding noun forms that are ambiguous between genitive-singular and nominative-plural. Slioussar asked whether the form ambiguity of such attractors as in (4a) might lead to increased production and acceptability of ungrammatical plural verbs, even when the syntactic context disambiguated them as genitive singular nouns such that the sentence thus contained no plural noun at all. She reported exactly this kind of error: when nominative-singular agreement controllers appeared with case+number syncretic genitive singular attractors, more plural verb agreement was erroneously produced and in comprehension plural verb agreement was erroneously considered acceptable. On the other hand, a similar study in Czech (Lacina and Chromỳ, 2022) showed a different pattern of results. Czech exhibits exactly the same accidental homophony between genitive singular and nominative plural forms as Russian. However, no effect of case+number syncretism was observed in Czech comprehension: reading times for ungrammatical plural verb sentences showed no difference for singular genitive attractors that were homophonous with the plural nominative vs. those that were not (Lacina and Chromỳ, 2022).

(4) a.

Komnata	dlja	večerinki	byli	...
room.NOM.SG	for	party.GEN.SG$_{(=\text{NOM.PL})}$	were	

‘The room for party were ...’

b.

Komnata	dlja	večerinok	byli	...
room.NOM.SG	for	party.GEN.PL$_{(\neq\text{NOM.PL})}$	were	

‘The room for parties were ...’

In a clever early production study, Bock and Eberhard (1993) tested a slightly different kind of form overlap effect on agreement attraction, by making use of singular-plural homophony. In English singular form *crew* and plural form *crews* are not number syncretic, but the plural form *crews* is homophonous with the singular form of another noun, *cruise*, what can be termed a pseudoplural. Bock and Eberhard tested whether pseudoplural singular attractors such as *cruise* as in (5a), increase agreement errors in the same way as true plural nouns, such as *crews* in (5b).

(5) a. The player on the cruise . . .

b. The player on the crews . . .

Participants in the study heard preambles as in (5a) or (5b) and were asked to repeat the preamble and complete the sentence. However, they found that pseudoplural attractors did not induce comparable agreement errors to true plural attractors.

The question of whether shared surface form or shared abstract specification drives interference is not unique to agreement. Similarity in case marking is known to affect the processing of other dependency configurations as well: multiply center-embedded sentences in Japanese, for example, become harder to process as the number of identically case-marked (nominative *-ga*) subjects increases (Uehara and Bradley, 1996; Uehara, 2003; Babyonyshev and Gibson, 1999). We return to this literature in the General Discussion, where we ask whether such case-similarity effects reflect the phonological form of the case markers or their abstract morphosyntactic specification.

1.3 Accounting for syncretism and phonological overlap

The form effects reviewed above matter because different explanations of them make different commitments about what information guides the parser's search for an agreement controller: form may influence agreement directly, by activating the features a form can express, or indirectly, by changing the estimated likelihood that the attractor is a controller. Several different accounts have been proposed to explain the observed effects of surface form on agreement attraction. One account from Slioussar (2018) proposes that phonological form itself mediates the search for the agreement controller. We will refer to this as the *phonological modulation* account, since on this view the surface form of the attractor modulates retrieval directly, without the mediation of any structural estimate. Situating her findings within cue-based retrieval accounts, Slioussar argued that the phonological form which overlaps between the genitive singular and nominative plural forms can activate relevant features like +NOM and +PL in addition to +GEN and +SG, which would increase the likelihood of retrieving the attractor as a controller, leading to more agreement errors in cases where the attractor only matches with the +GEN and +PL cues (4a). In other words, a phonological form just activates, in a context-independent way, all the feature sets that the form can be used to express in the language. To the best of our knowledge, this is the only account that directly exploits the co-activation of chunks through phonological overlap to explain the effects of syncretism. However, this account faces several empirical challenges. First, if phonological forms activate all features compatible with the form, independent of context, this would seem to predict that phonologically ambiguous singular-plural forms in English as in (5a) should lead to plural attraction at the verb, contrary to what Bock and Eberhard (1993) found. Second, this account would predict that case+number syncretic forms such as those tested by Slioussar (2018) should lead to attraction independent of other properties of the language, but Lacina and Chromỳ (2022) showed that this does not occur in Czech.

A second family of accounts posits a more indirect relation between surface form and agreement processes. In these accounts, the effect of form is mediated by its influence on the estimate of structural properties related to agreement–either the likelihood that a noun is a subject, or the likelihood that a noun is an agreement controller. Comprehenders may probabilistically associate certain surface cues—word order, case morphology, or the presence of certain morphemes—with subjecthood or controllerhood, based on the distribution of these forms in the language. Case forms that are syncretic with the nominative would then carry an increased association with subjecthood or controllerhood. For example, Lago et al. (2019) argue that Turkish speakers retrieve genitive-marked attractors as controllers because genitive case controls agreement in embedded clauses, even though it cannot do so in matrix clauses. The genitive possessor thus shares its surface form with a genuine agreement controller (the embedded subject) without sharing its controller function in the sentence at hand, and attraction arises because Turkish speakers associate genitive-marked nominals with being controllers.

This account offers a potential resolution to the Russian–Czech puzzle. While Russian genitive-marked nouns can serve as subjects in negative inversion constructions, they do not control verbal agreement in these contexts. Importantly, however, they remain active controllers within the noun phrase, triggering number or gender marking on modifiers (e.g., surfacing as feminine *ni odnoy* with a feminine head, contrasting with masculine *ni odnogo* in 6) (Babby, 2001; Partee and Borschev, 2004). Genitive marking in Russian is therefore weakly but genuinely associated with controllerhood. In contrast, Czech genitives only have the second property–they control concord on their modifiers just as Russian ones do—but not the first: genitive subjects in negative contexts are unavailable. Therefore, Russian and Czech is distinguished by whether a genitive subject can have association with being a subject and a controller. Only in Russian does genitive marking carry an association, however weak, with nominals that occupy the subject position from which verbal agreement is controlled; in Czech, that association is absent, and the same surface homophony fails to produce attraction.

(6)

Tam	ne	rabotaet	ni	odnogo	inženera.
there	NEG	works	not	one.M.SG.GEN	engineer.M.SG.GEN

'There hasn't been a single engineer working there.'

Converging evidence for this sensitivity to "looking like a controller" comes from Romanian (Bleotu and Dillon, 2024). Bleotu and Dillon (2024) found that Romanian attractors induced agreement errors only when they surfaced with a determiner, as opposed to bare forms. Since only nouns bearing a determiner can control agreement in Romanian, they argue that participants associate the presence of a determiner with controllerhood.

Taken together, these explanations across languages and structures suggest that the cue–chunk match is not strictly categorical but can be influenced by statistical associations between surface forms and structural roles within a language (Engelmann et al., 2019). We return to further evidence for this view from Hindi and English in the General Discussion.

1.4 Turkish grammar sketch

This section provides a brief sketch of the properties of Turkish grammar that our experiments build on: basic clause structure and number agreement, the genitive-possessive construction, which served as the attractor in previous Turkish studies and supplies our baseline nominal attractors, and nominalized relative clauses, which supply our verbal attractors.

Turkish is a head-final (SOV), agglutinating language: grammatical functions are signaled by case suffixes on noun phrases, and subject–verb agreement is marked on the verb by person/number suffixes. Number agreement is asymmetric. A singular subject cannot combine with a plural-marked verb (7b), whereas with a plural subject the plural agreement suffix *-lAr* is optional (7c) (Göksel and Kerslake, 2005). As (7c) also shows, the same suffix *-lAr* serves both as plural marking on nouns (*çocuk-lar* 'kids') and as plural agreement on verbs (*ko-tu-lar* 'they ran'); this dual role is central to our manipulation.

(7) a. Çocuk ko-tu.
kid[NOM] run-PST
'The kid ran.'

b. * Çocuk ko-tu-lar.
kid[NOM] run-PST-PL
Intended: 'The kid ran.'

c. Çocuk-lar ko-tu-(lar).
kid-PL run-PST-(PL)
'The kids ran.'

Similar to English Saxon genitives, in Turkish possessive constructions the possessor is overtly marked with genitive case, and the possessed head bears a possessive suffix, as in (8). Noun phrases of this type served as the attractors in previous Turkish attraction studies, and they constitute the baseline nominal attractors in our Experiment 1.

(8) Milyoner-ler-in aç-s
millionaire-PL-GEN cook-POSS
'the millionaires' cook'

Genitive case does more than mark possessors. The most common way of embedding a clause in Turkish is nominalization: the embedded verb bears a nominalizing suffix such as *-DIK*, the embedded subject appears in the genitive, and the resulting clause is case-marked like a noun phrase, as in (9). Importantly, unlike English nominalizations such as *the kids' arrival*, the nominalized verb in Turkish agrees in number with its genitive subject: in (9), plural *-lAr* on the genitive subject is cross-referenced by *-lAr* on the nominalized verb. A genitive-marked noun phrase in Turkish is therefore a possible subject within the language and can control the agreement.

(9) Çocuk-lar-ın gel-dik-ler-in-i duy-du-m.
Kid-PL-GEN arrive-NMLZ-PL-POSS-ACC hear-PST-1SG.
'I heard kids' arrival.'

Relative clauses in Turkish are built from the same nominalizations and precede their head. In a non-subject relative clause, the verb bears *-DIK* and possessive agreement with its genitive subject, as in *öretmen-in gör-dü-ü çocuk* (teacher-GEN see-NMLZ-POSS kid) 'the kid that the teacher saw'. Because Turkish freely drops arguments available in context, the genitive subject can be omitted, yielding a reduced relative clause (*gör-dü-ü çocuk* 'the kid that (s/he) saw'). The head noun can be dropped as well, yielding a headless relative clause (*gör-dü-ü* 'the one that (s/he) saw'), which can occupy any argument position, including subject position. When plural marking enters these forms, the string becomes ambiguous: in *gör-dük-ler-i*, the surface string alone does not determine whether *-lAr* marks the plurality of the entities seen or verbal agreement indicating that the (unexpressed) subject of the relative clause is plural.

The agreement behavior of these plural-marked forms is the crucial fact for our design. Consider a context in which ten kids were playing outside, and the teacher managed to see only three of them. In this context, an overt plural head noun controls agreement on the matrix verb, with the usual optionality of plural marking (10a). A headless relative clause can also stand as the matrix subject and refer to the same three kids (10b). Even so, plural agreement on the matrix verb is marked and rarely used (10c): although the subject string bears *-lAr* and refers to a plurality, verbal *-lAr* does not ordinarily control agreement. Strings bearing verbal *-lAr* can thus surface as subjects, but their use as a controller of matrix verb agreement is marked.

(10) a. Gör-dü-ü çocuk-lar ko-tu-(lar).
see-NMLZ-POSS kid-PL run-PST-(PL)
'The kids that (s/he) saw ran.'

b. Gör-dük-ler-i ko-tu.
see-NMLZ-PL-POSS run-PST
'The ones that (s/he) saw ran.'

c. ? Gör-dük-ler-i ko-tu-lar.
see-NMLZ-PL-POSS run-PST-PL
Intended: 'The ones that (s/he) saw ran.'

Although we refer to these *-DIK* forms as verbal attractors, they are nominalizations, and they pattern with noun phrases in their external syntax: they bear case and possessive morphology, occupy argument positions, and can be coordinated like noun phrases (Göksel and Kerslake, 2005; Kornfilt, 1984). When they modify a head noun—as in our materials—they are analyzed as entity-denoting rather than propositional (Demirok, 2019). What distinguishes them from genitive nominal attractors is thus not their syntactic distribution but their agreement profile: genitive nominals control agreement in the constructions where they act as subjects, whereas forms bearing verbal *-lAr* never do. We return to the categorial status of these forms in the General Discussion.

1.5 The present study

Motivated by these alternative accounts and conflicting findings (Bock and Eberhard, 1993; Slioussar, 2018; Lacina and Chromỳ, 2022), we test the phonological modulation account in three experiments: whether a syntactically ineligible controller, but still a possible subject, can induce attraction solely through morphophonological overlap with the agreement suffix. To this end we capitalize on the dual role of *-lAr* described in the previous section. Our attractors are reduced relative clauses (RRCs) in which the plural-marked nominalized verb modifies a singular head noun, the true agreement controller (11).

(11) Gör-dük-ler-i çocuk ko-tu-(*lar).
see-NMLZ-PL-POSS kid[NOM] run-PST-(*PL)
'The kid that (they) saw ran.'

As the grammar sketch makes clear, verbal *-lAr* presents the reverse image of the Czech genitive. Czech genitive nominals cannot surface as subjects, although they control concord within their noun phrase; Turkish strings bearing verbal *-lAr* can surface as subjects, as headless relative clauses do, but their use as controllers of verbal agreement is marked and rare (10c). Like Czech, and unlike Russian, the attractor's form is therefore not associated with controllerhood. The two accounts introduced in §1.3 thus make opposite predictions for our materials. The controllerhood-association account predicts no attraction: verbal *-lAr* is not associated with agreement controllers, so it carries no controller-relevant cue. The phonological modulation account, by contrast, predicts attraction: the surface form *-lAr* can activate both the verbal and the nominal number specifications compatible with it, so the attractor's chunk can acquire a spurious plural cue-match through surface overlap alone.

These predictions concern ungrammatical sentences, where attraction is most reliably observed. We make no specific prediction about attraction in grammatical sentences: as Hammerly et al. (2019) argue, the apparent asymmetry of attraction between grammatical and ungrammatical sentences can be understood as a by-product of response bias rather than of the interference process itself. Response bias does, however, license one prediction of its own: wherever attraction is genuinely present, it should be visible even in participants who show no bias toward 'yes' responses. We take up this point in detail in §5.4.

In Experiment 1, we tested these predictions by comparing sentences with verbal attractors to sentences with canonical nominal attractors in Turkish. Experiment 2 then tested the phonological modulation account more directly by using only verbal attractors. Finally, Experiment 3 tested retrieval in real time, using the verbal attractor conditions in self-paced reading. Attraction from verbal attractors—higher acceptance of ungrammatical plural verbs in judgments, and a reduced ungrammaticality penalty in reading times—would support the phonological modulation account; its absence, alongside robust attraction from nominal attractors, would support the controllerhood-association account.

Across three experiments, we found no evidence that verbal *-lAr* induces attraction; in Experiment 1, this held even when robustly attracting nominal items appeared in the same session. This pattern aligns with prior findings in the attraction literature and in Turkish specifically: surface-form overlap alone does not give rise to agreement illusions. Rather, attraction appears to depend on abstract feature overlap between potential controllers and agreement probes, and possibly on statistical associations between surface forms and controllerhood. In this light, the findings of Slioussar (2018) are best analyzed as reflecting an association between Russian genitive marking and controllerhood, an association that Czech genitive marking lacks; the absence of attraction in

Czech follows (Lacina and Chromỳ, 2022). Together, the three experiments clarify which cues guide retrieval and delimit the role of phonological overlap in dependency resolution.

2 Experiment 1: Speeded acceptability with nominal and verbal attractors

Previous studies have established robust agreement attraction in Turkish speeded acceptability judgments with plural genitive possessors (Lago et al., 2019; Türk and Logačev, 2024): participants accept ungrammatical plural agreement on the matrix verb more often when a genitive-marked possessor inside the subject phrase is plural than when it is singular. Experiment 1 uses this configuration as a within-experiment benchmark. Nominal attractor conditions adapted from Türk and Logačev (2024) are compared with verbal attractor conditions in which the same plural morpheme *-lAr* appears on a nominalized verb. If surface-form overlap alone drives attraction, the two attractor types should produce comparable attraction profiles.

2.1 Participants

We recruited 95 undergraduate students to participate in the experiment in exchange for course credit. All participants self-identified as native Turkish speakers, with an average age of 21 years (range: 18-30). The experiment was carried out following the principles of the Declaration of Helsinki and the regulations concerning research ethics at Boğaziçi University. All participants provided informed consent before their participation and their identities were completely anonymized.

2.2 Materials

We used 40 sets of sentences like Table 1, in which we manipulated (i) the number of the attractor, (ii) the type of the attractor, and (iii) the number agreement on the verb. Both plural markings were marked with the suffix *-lAr*, while the singular number and singular agreement were marked by its absence.

Table 1: Experimental conditions. The Attractor was manipulated for number and type. The Verb was also manipulated for number to match or mismatch the head noun (always singular), creating Grammatical (SG) and Ungrammatical (PL) conditions.

Att. Type	**Att. Num.**	**Verb Num.**	**Sentence & Gloss**				
Verbal	**SG**	**SG**	Tut-tu-u	aç	mutfak-ta	sürekli	zpla-d
		Grammatical	*hire*-NMLZ-POSS	*cook*	*kitchen*-LOC	*non.stop*	*jump*-PST
		PL	*Tut-tu-u	aç	mutfak-ta	sürekli	zpla-d-lar
		Ungrammatical	*hire*-NMLZ-POSS	*cook*	*kitchen*-LOC	*non.stop*	*jump*-PST-PL
			'The hired$_{sg}$ cook jumped in the kitchen non-stop.'				
Verbal	**PL**	**SG**	Tut-tuk-lar-	aç	mutfak-ta	sürekli	zpla-d
		Grammatical	*hire*-NMLZ-PL-POSS	*cook*	*kitchen*-LOC	*non.stop*	*jump*-PST
		PL	*Tut-tuk-lar-	aç	mutfak-ta	sürekli	zpla-d-lar
		Ungrammatical	*hire*-NMLZ-PL-POSS	*cook*	*kitchen*-LOC	*non.stop*	*jump*-PST-PL
			'The hired$_{pl}$ cook jumped in the kitchen non-stop.'				
Nominal	**SG**	**SG**	Milyoner-in	aç-s	mutfak-ta	sürekli	zpla-d
		Grammatical	*millionaire*-GEN	*cook*-POSS	*kitchen*-LOC	*non.stop*	*jump*-PST
		PL	*Milyoner-in	aç-s	mutfak-ta	sürekli	zpla-d-lar
		Ungrammatical	*millionaire*-GEN	*cook*-POSS	*kitchen*-LOC	*non.stop*	*jump*-PST-PL
			'The millionaire's cook jumped in the kitchen non-stop.'				
Nominal	**PL**	**SG**	Milyoner-ler-in	aç-s	mutfak-ta	sürekli	zpla-d
		Grammatical	*millionaire*-PL-GEN	*cook*-POSS	*kitchen*-LOC	*non.stop*	*jump*-PST
		PL	*Milyoner-ler-in	aç-s	mutfak-ta	sürekli	zpla-d-lar
		Ungrammatical	*millionaire*-PL-GEN	*cook*-POSS	*kitchen*-LOC	*non.stop*	*jump*-PST-PL
			'The millionaires' cook jumped in the kitchen non-stop.'				

Verbal attractor conditions featured complex subject NPs containing a bare head noun and a reduced relative clause acting as the attractor (e.g., '*tuttuklar aç*' 'the hired cook'). Because nominal plural marking is mandatory and the head noun was always singular, plural verb agreement rendered these sentences ungrammatical. Nominal attractor conditions, featuring nominal attractors such as '*milyonerlerin açs*' ('the millionaires' cook') were taken from Türk and Logačev (2024).[2] To prevent participants from associating plural verbs with ungrammaticality, fillers were

[2]All head nouns in these materials are vowel-final, so that the third person singular possessive surfaces as *-sI* and

balanced between grammatical sentences with plural verbs and ungrammatical sentences with singular verbs.

2.3 Procedures

The experiment was conducted online via Ibex Farm (Drummond, 2013), lasting approximately 25 minutes. After providing informed consent and demographic details, participants read instructions and completed nine practice trials.

Each trial began with a 600 ms blank screen, followed by a centered, word-by-word RSVP presentation (30 pt font, 400 ms duration, 100 ms inter-stimulus interval). Upon the prompt, participants judged sentence acceptability as quickly as possible by pressing 'P' (acceptable) or 'Q' (unacceptable). A red warning message appeared during practice trials—but not experimental trials–if responses exceeded 5,000 ms. Participants pressed the space bar to advance to the next item.

The study included 40 experimental and 40 filler sentences. Experimental items were distributed across eight lists using a Latin-square design, ensuring each participant viewed only one list containing one version of each item.

2.4 Analysis

Prior to the analyses, we excluded participants whose 'yes' response rate difference between grammatical (singular-verb) and ungrammatical (plural-verb) sentences with singular attractors was below 25%, applying this criterion separately to the nominal and the verbal conditions. One participant was below threshold in the nominal conditions and two in the verbal conditions, resulting in

is form-wise distinct from the accusative *-(y)I*. While a reader familiar with the Turkish processing literature may take this property to instantiate the disambiguation strategy of Dinçtopal-Deniz (2014) and Deniz and Fodor (2017), the two manipulations run in opposite directions. Those studies make use of the string ambiguity of *-I*: with the consonant-final head nouns they employ, as in Lago et al. (2019), the possessive and the accusative are homophonous, and the ambiguity is resolved by the following syntactic and prosodic context. The vowel-final endings adopted by Türk and Logačev (2024), and inherited here, instead removes that ambiguity, so that a temporary analysis of the genitive phrase as a clause-level subject and ambiguously marked attractor as the object is unavailable. The allomorphy itself is long established in descriptive grammars of Turkish (Lewis, 2000; Kornfilt, 1997).

3 unique participants excluded at subject level (3.16% of the recruited sample). We additionally removed trials with too slow or too fast responses (111 with RT < 200 ms; 60 with RT > 4999 ms; 2.32% of trials). After Response Time trimming, there were no missing responses. In total, 5.41% of collected trials were excluded, and the final dataset contained 92 participants and 7189 observations.

All analyses were conducted in R using the `brms` package (Bürkner, 2021) as an interface to Stan for Bayesian hierarchical modeling (Stan Development Team, 2026), with tidyverse for data preparation and `tidybayes/Rmisc` for posterior summaries and within-subject error computation (Wickham et al., 2019; Kay, 2024; Hope, 2022). A Bayesian hierarchical model assuming the Bernoulli distribution (logit link) was fit to the binary acceptability response, with the dependent variable coded as 1 for an 'acceptable' rating and 0 otherwise. Grammaticality, Attractor Number, and their interaction were included as fixed effects. The random-effects structure was maximal: by-subject and by-item random intercepts and random slopes for Grammaticality, Attractor Number, and their interaction, including correlations (Barr et al., 2013). Following Avetisyan et al. (2020), we used weakly informative priors for fixed effects ($\mathcal{N}(0, 1)$) and standard regularizing priors for remaining parameters. The $\mathcal{N}(0.85, 0.70)$ prior is on the intercept parameter (log-odds scale), not directly on the observed data. Its central 95% prior interval is approximately [-0.52, 2.22], corresponding to a baseline *yes* probability of about [0.37, 0.90]. Since we used sum contrasts as in Table 3, the intercept refers to an overall average of all factors, i.e. when the coefficient is 0. Full model specifications and priors are reported in Table 2.

Through out the paper, descriptive term CI denotes a 95% confidence interval around a descriptive mean (Clopper–Pearson intervals for the binomial proportions shown in figures), whereas CrI denotes a 95% Bayesian credible interval around a posterior estimate. Reading-time figures in Experiment 3 show within-subject 95% confidence intervals computed with the method of Morey and Cousineau (Morey, 2008; Cousineau, 2005).

Table 2: Bayesian model specifications for Experiment 1.

Specification	Value	Sampling	Value
Family (link)	Bernoulli (logit)	Chains	4
Intercept prior	$\mathcal{N}(0.85, 0.70)$	Iterations	12,000 (2,000 warmup)
Fixed effects	$\mathcal{N}(0, 1)$	Threads	4
Random-effect SD	Exponential(1)	`sample_prior`	yes
Random-effect corr.	LKJ(2)	Init / seed	0 / 1

Table 3: Contrast coding used in the Experiment 1 Bayesian model. All predictors are sum-coded.

Factor	+0.5	−0.5
Grammaticality (G)	Grammatical	Ungrammatical
Attractor Number (A)	Plural	Singular
Attractor Type (T)	Verbal	Nominal

2.5 Results

Participants showed high accuracy on both grammatical fillers, which they accepted at a rate of M = 0.95 (95% CI = [0.94,0.96]), and ungrammatical fillers, which they correctly rejected at a rate of M = 0.94 (95% CI = [0.93,0.95]), indicating that they understood the task and performed it reliably.

Figure 1 shows mean 'yes' responses with 95% confidence intervals across conditions. In the nominal-attractor conditions, we see the expected attraction profile: ungrammatical sentences with plural genitive attractors received more 'yes' responses (M = 0.11, CI = [0.09,0.15]) than ungrammatical sentences with singular genitive attractors (M = 0.04, CI = [0.03,0.07]). Within grammatical sentences, we see the reversed trend of plural attractor decreased 'yes' responses, though overlapping confidence intervals. Participants accepted singular attractor more often (M = 0.91 CI = [0.88,0.94]) compared to their plural attractor counterparts (M = 0.87 CI = [0.84,0.90]).

This modulation by the attractor number was not observed with verbal relative-clause attractors. Ungrammatical verbal-attractor items remain near floor in both number conditions (singular: M = 0.04, CI = [0.02, 0.06]; plural: M = 0.04, CI = [0.03, 0.06]). Similarly, grammatical sentences were generally rated highly across attractor number (singular: M = 0.94, CI = [0.92, 0.96]; plural:

M = 0.94, CI = [0.91, 0.96]).

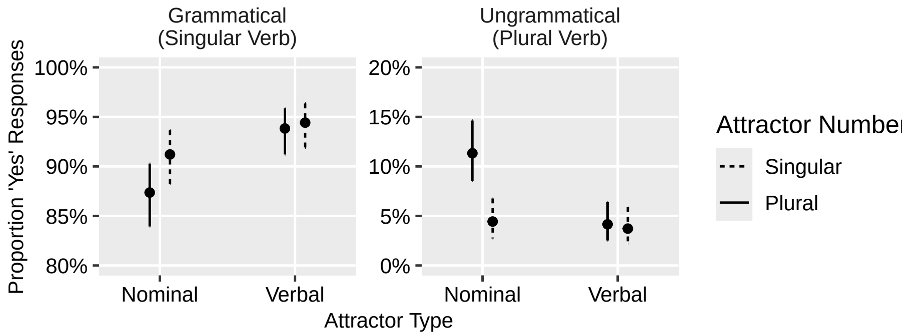


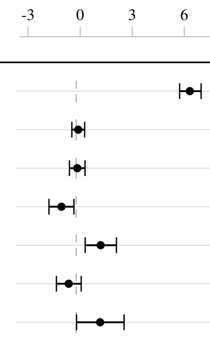
Figure 1: Mean proportion of 'acceptable' responses by grammaticality, attractor number and attractor type. Error bars show 95% Clopper–Pearson confidence intervals.

Table 4 reports posterior summaries for fixed effects in the our main model. The model shows a strong grammaticality effect (G: 6.55 [5.96, 7.20], P(β>0) >0.99), with grammatical sentences were given 'yes' responses far more often than ungrammatical ones. The critical attraction term is negative, indicating a reduced grammaticality effect under plural attractors, and its credible interval excludes zero (G × A: −0.85 [−1.57, −0.13], P(β>0) = 0.01). Most importantly for our hypothesis, this attraction term is modulated by attractor type as shown by a positive and credible three way interaction (G × A × T: 1.38 [0.02, 2.76], P(β>0) = 0.98). Because verbal attractors are coded positively, the positive three-way term indicates that attraction seen in the two interaction is concentrated in the nominal conditions and largely absent in the verbal attractor conditions. The model also shows a smaller grammaticality effect for nominals than for verbal attractor items (G × T: 1.41 [0.52, 2.32], P(β>0) >0.99), possibly driven by the increased acceptability in ungrammatical sentences due attraction. Neither main effect of attractor number (0.12 [−0.25, 0.49]) nor of attractor type (0.06 [−0.39, 0.51]) is credibly different from zero.

To directly test whether attraction is present in verbal conditions, we fit a Bayesian model with identical specifications restricted to the verbal-attractor trials. In this follow-up model, the Grammaticality - Attractor Number interaction was −0.18 [−1.36, 1.02] (P(β>0) = 0.37). Because the posterior interval spans zero and the directional probability is not extreme, we conclude that we do not have any evidence for attraction effects within verbal conditions in Experiment 1.

Table 4: Posterior summaries of key fixed effects in the full Bayesian model for Experiment 1. All predictors were sum-coded (±0.5). Whisker plots show the posterior mean and 95% CrI on a common scale from -3.5 to 7.8; the dashed line marks zero.

Effect	β	95% CrI	$P(\beta > 0)$
Grammaticality (G): Gram vs. Ungram	6.55	[5.96 , 7.20]	> 0.99
Attractor Number (A): Pl vs. Sg	0.12	[−0.25 , 0.49]	0.75
Attractor Type (T): Verbal vs. Nominal	0.06	[−0.39 , 0.51]	0.60
G × A	−0.85	[−1.57 , −0.13]	0.01
G × T	1.41	[0.52 , 2.32]	> 0.99
A × T	−0.43	[−1.14 , 0.29]	0.12
G × A × T	1.38	[0.02 , 2.76]	0.98



To quantify evidence for the null effect in the verbal conditions, we compared two verbal-condition-only Bayesian mixed-effects models using Bayes factors estimated by bridge sampling in `brms` (Bürkner, 2021). We compared two models: the model (M_{int}) that included the critical Grammaticality - Attractor Number interaction and the null model ($M_{\emptyset}$) that omitted it while keeping the same main effects and random-effects structure, so the comparison directly tests whether verbal attractors induce agreement attraction. We report $BF_{01} = P(Data|M_{\emptyset}) / P(Data|M_{int})$, which under equal prior odds expresses how much more likely the observed data are under the null model. Following Jeffreys (1961, as cited in Lee and Wagenmakers (2014)), we treat values of 3–10 as moderate evidence, 10–30 as strong, 30–100 as very strong, and above 100 as extreme.

Because Bayes factors depend on the prior assigned to the parameter of interest, we refit the same pair of models under a range of increasingly concentrated zero-centered normal priors on the fixed effects (Table 5). BF_{01} favors the null model under every prior, ranging from 2.73 to 12.40, so the direction of the conclusion does not depend on allowing large effects a priori. Its strength does vary, from anecdotal under the narrowest priors to strong under the widest, and we therefore read the result as consistent evidence against verbal attraction.

To interpret this pattern at the condition level and situate them in general Turkish attraction effects, we fitted a further model that also included the data of Türk and Logačev (2024). Contrast coding followed the models above, except that attractor type now had three levels (verbal [V], Nominal from our experiment [NOM.CUR], Nominal from TL24[NOM.TL24]) and was Helmert-coded:

Table 5: Prior-sensitivity analysis for the RC-only Bayes factor comparison. $BF_{01} > 1$ favors the model without the Grammaticality x Attractor Number interaction ($M_{\emptyset}$).

Prior set	Prior on fixed effects	BF_{01}	Evidence against interaction
Wide	$\mathcal{N}(0, 2.00)$	12.40	strong
Main	$\mathcal{N}(0, 1.00)$	4.73	moderate
Narrow	$\mathcal{N}(0, 0.50)$	4.60	moderate
Very narrow	$\mathcal{N}(0, 0.25)$	2.73	anecdotal
Ultra narrow	$\mathcal{N}(0, 0.10)$	3.00	anecdotal

`T1` contrasts the V condition against the mean of the two nominal conditions (V = 0.333, NOM.CUR = NOM.TL24 = −0.167), and `T2` contrasts the two nominal conditions with each other (NOM.CUR = 0.333, NOM.TL24 = −0.333, V = 0). Combining the posterior draws for `G` × `A` with those for `G` × `A` × `T1` and `G` × `A` × `T2` according to these weights recovers the model-implied attraction effect within each condition; we summarize each by its posterior mean, 95% credible interval, and P(β<0), with more negative values indicating stronger attraction or acceptability difference. Figure 2 summarizes these derived contrasts. Direct model coefficients for all of our additional models in this section can be found in A.

Both nominal conditions show clear attraction (NOM.CUR: −1.59 [−2.43, −0.77]; NOM.TL24: −1.17 [−1.73, −0.59]; both P(β<0) >0.99), whereas the verbal attractor condition shows little or none (−0.18 [−1.18, 0.83], P(β<0) = 0.63), and the two nominal datasets do not differ credibly from each other (−0.64 [−2.14, 0.85], P(β<0) = 0.80) with respect to the magnitude of attraction. The main effect of `T2` in the same model, however, indicates lower baseline acceptability in the current nominal dataset than in Türk and Logačev (2024) (−1.10 [−1.77, −0.43], P(β<0) >0.99), possibly due to a shift in baseline acceptability within nominal acceptability without a corresponding shift in attraction magnitude.

2.6 Discussion

Experiment 1 tested whether phonological overlap between nominal and verbal plural marking in Turkish is sufficient to induce agreement attraction. Our results did not show evidence for verbal

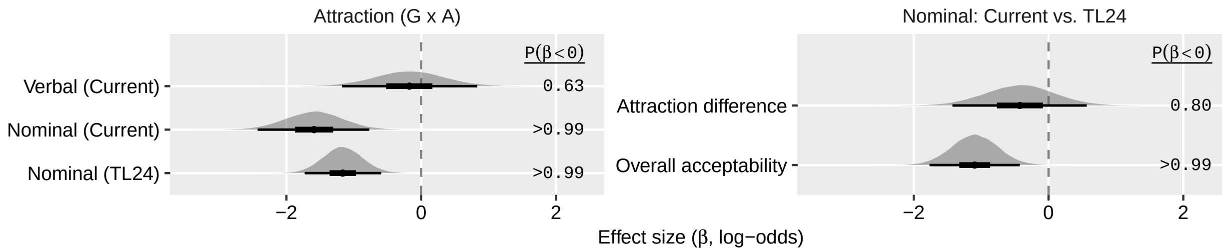


Figure 2: Posterior summaries of attraction-related effects. Points indicate posterior means, and horizontal bars show 95% credible intervals on the log-odds (β) scale. Attraction was estimated as the interaction between grammaticality and attractor number within each attractor type. Negative values indicate stronger attraction (a reduced ungrammaticality penalty in plural-attractor conditions). Dashed line denotes zero (no effect).

attraction. Participants reliably rejected ungrammatical sentences with plural-marked verbal attractors, while showing canonical attraction in the nominal-attractor conditions that replicate prior work (Türk and Logačev, 2024; Lago et al., 2019). We confirmed that this dissociation is not an artifact of the mixed design: a follow-up model restricted to verbal-attractor trials showed no interaction between Grammaticality and Attractor Number. Our Bayes Factor and prior sensitivity analysis also suggested that the models without the interactions were preferred over the models with interaction included. This pattern is more consistent with accounts in which attraction depends on controller-relevant cues rather than surface-form overlap alone (Lago et al., 2019); we return to this in the General Discussion.

A potential concern is the mixed design, which combined canonical nominal attractors with verbal ones. The presence of robustly attracting nominal items could have shifted participants' response strategies, either masking a weaker verbal effect or encouraging direct comparison between the two attractor types (Hammerly et al., 2019; Arehalli and Wittenberg, 2021; Türk, 2022). Although we found no such effect on attraction in the nominal conditions, overall acceptability for nominal items was lower in the present study than in Türk and Logačev (2024), suggesting that the experimental context might have modulated response patterns, thus attraction effects in verbal conditions (we discuss the implications of this distributional shift further in the General Discussion). To determine whether the absence of attraction in verbal condition is genuine or a distributional artifact of the mixed design, Experiment 2 removed all nominal attractors, testing whether the null effect persists when verbal morphology is the only possible attractor between trials.

3 Experiment 2: Verbal attractors in isolation

Experiment 1 showed no attraction from verbal attractors, but the presence of robust nominal-attractor items may have influenced response patterns (Hammerly et al., 2019; Arehalli and Wittenberg, 2021; Türk, 2022). To address a possible effect of response patterns, we conducted Experiment 2 , removing all nominal-attractor conditions and testing whether the null effect persists when verbal conditions do not have a possible point of comparison with robust nominal attractors.

3.1 Participants

We recruited 80 undergraduate students to participate in the experiment in exchange for course credit. All participants self-identified as native Turkish speakers, with an average age of 21 years (range: 18-31). The experiment was carried out following the principles of the Declaration of Helsinki and the regulations concerning research ethics at Boğaziçi University. All participants provided informed consent before their participation and their identities were completely anonymized.

3.2 Materials

We used the same verbal attractor items and fillers as in Experiment 1, removing all nominal attractor conditions.

3.3 Procedures

The procedure was identical to Experiment 1. However, since the condition number was reduced to four, experimental items were distributed across four lists using a Latin-square design, ensuring each participant viewed only one list containing one version of each item.

3.4 Analysis

Prior to the analyses, we excluded participants whose 'yes' response rate difference between grammatical (singular-verb) and ungrammatical (plural-verb) sentences with singular attractors was below 25%. This removed 2 participant(s) (2.5% of the recruited sample). We additionally removed trials with too slow or too fast responses (154 with RT $<$ 200 ms; 31 with RT $>$ 4999 ms; 2.96% of trials). After Response Time trimming, there were no missing responses. In total, 5.39% of collected trials were excluded, and the final dataset contained 78 participants and 6055 observations. All other analysis details were the same as in Experiment 1.

3.5 Results

Participants showed high accuracy in both grammatical (M = 0.94, CI = [0.92,0.95]) and ungrammatical filler sentences (M = 0.92, CI = [0.90,0.93]), indicating that they understood the task and performed it reliably.

Figure 3A and B present the overall means and credible intervals for 'yes' (acceptable) responses across experimental conditions, along with posterior estimates from our model. As shown, ungrammatical sentences with plural attractors were rated as acceptable as their counterparts with singular attractors (M = 0.06 and 0.05, CI = [0.04, 0.07] and [0.03, 0.07] for singular and plural attractors, respectively).

On the other hand, accuracy in grammatical conditions was modulated by the number of the attractor in an unexpected way. Participants rated grammatical sentences with singular attractors as grammatical less often (M = 0.92, CI = [0.9,0.94]) compared to their counterparts with plural attractors (M = 0.95, CI = [0.93,0.96]).

We fitted a Bayesian mixed-effects logistic regression to binary *yes/no* responses using `brms`, with the same model family (Bernoulli, logit link) as in Experiment 1. As there, the priors were weakly informative: $\mathcal{N}(0, 1)$ on the fixed effects, Exponential(1) on the random-effect standard deviations, and LKJ(2) on the random-effect correlations, with an $\mathcal{N}(0.85, 0.70)$ prior on the intercept. Because Experiment 2 included only verbal attractor conditions, the model had a simpler 2 $\times$

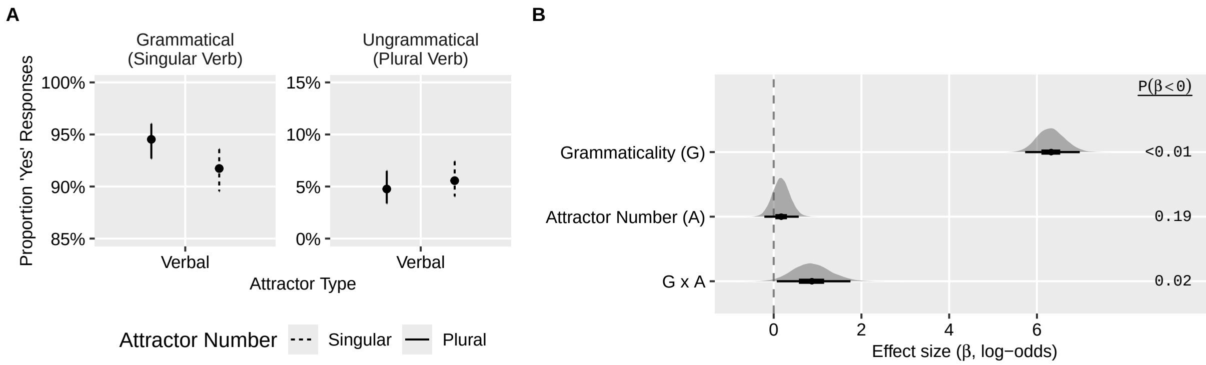


Figure 3: A: Mean proportion of ‘acceptable’ responses by grammaticality and attractor number. Error bars show 95% Clopper–Pearson confidence intervals. B: Posterior means and 95% credible intervals for fixed effects in the Experiment 2 Bayesian model (log-odds scale).

2 structure: fixed effects for Grammaticality and Attractor Number (both sum-coded at ±0.5) and their interaction, with maximal random intercepts and slopes for both subjects and items.

The model revealed a positive effect of grammaticality (β = 6.32 [5.73, 6.98], P(β>0) <0.01), but no reliable main effect of attractor number (β = 0.17 [-0.21, 0.57], P(β<0) = .19). We also found substantial evidence for a small but positive interaction (β = 0.87 [0.07, 1.75], P(β > 0) = .98). This interaction reflected a aforementioned asymmetry in the grammatical-condition rather than genuine attraction. To clarify the pattern within each grammaticality level, we computed model-implied attractor-number effects from posterior draws. In grammatical conditions, plural attractors increased acceptance relative to singular attractors (β = 0.61 [0.06, 1.20], P(β>0) = 0.99), which cannot be explained via attraction mechanism since we would expect reduced acceptability with plural attractors. In ungrammatical conditions, there was no evidence for an attractor-number effect (β = -0.26 [-0.87, 0.31], P(β>0) = 0.19). The presence of a plural verbal attractor did not increase the acceptability of ungrammatical sentences. Thus, we conclude that the positive interaction is driven entirely by the pattern in the grammatical-condition that is opposite in direction to what an attraction account would predict.

3.6 Discussion

Experiment 2 replicated the verbal attractor conditions from Experiment 1 in isolation and again revealed no evidence for agreement attraction driven by plural markers on the verbal attractor. Ungrammatical sentences with reduced clause verbs serving as verbal attractors were rejected at similar rates regardless of attractor number. This confirms that the absence of verbal attraction in Experiment 1 was not due to the presence of nominal attractor items or to the distributional properties of the mixed design.

Unexpectedly, grammatical sentences with singular attractors were judged less acceptable than those with plural attractors. This effect is unlikely to reflect agreement attraction, since it arises in the opposite direction from what attraction would predict: grammatical sentences with singular attractors were rejected more often than those with plural attractors. One possibility is that it results from an interaction between plausibility and referential availability: the plural morpheme can license a more general interpretation, whereas the singular reduced relative clause more strongly invites a specific referent that may be less accessible in the task context. Alternatively, reduced relative clauses without a head may receive a correlative parse in Turkish: *gör-dü-ü* (see-NMLZ-POSS) can be interpreted as 'who(ever) he saw', and although this reading is notionally plural, correlatives of this kind trigger singular agreement in Turkish. It is therefore possible that notional plurality gave rise to a *reverse* attraction effect. We return to the possible explanations for this pattern, as well as the theoretical implications of the converging null result across both experiments, in the General Discussion.

4 Experiment 3: Verbal attractors in self-paced reading

Experiments 1 and 2 showed that verbal attractors do not induce agreement attraction in Turkish, despite carrying the same morphophonological marking as nominal attractors and being able to stand as subjects on their own. Both experiments used speeded acceptability judgments, which record the outcome of comprehension rather than its time course, and therefore provide limited insight into retrieval interference as it unfolds. Experiment 3 tested the same verbal attractor conditions in self-paced reading.

4.1 Participants

We recruited 79 monolingual native Turkish speakers (mean age 29 years, range 18–55) through Boğaziçi University mailing lists. Participants were paid at an hourly rate of approximately 12 USD; since the experiment took 25 minutes, they received 250 TRY (5.30 USD). The experiment was carried out following the principles of the Declaration of Helsinki and the regulations concerning research ethics at the University of Maryland, College Park. All participants provided informed consent before their participation and their identities were completely anonymized.

4.2 Materials

The experimental sentences were based on those of Experiment 2. Because the critical verb was the last word of the sentence in the earlier items, we embedded the verbal attractor configuration under matrix verbs such as *duy-* (hear), *oku-* (read), and *bil-* (know), with the overt complementizer *diye* (that), so that reading times at the critical verb would not be confounded with sentence wrap-up. To reduce interference from a further noun phrase, the matrix subject was pro-dropped and the matrix verb carried first-person marking. We also added an adjunct in sentence-initial position, so that the attractor was not the first element participants read and reading times at the attractor and the head, as well as their salience in the memory, were not affected by sentence-onset effects. The rest of the sentence was kept the same as in Experiments 1 and 2. Table 6 gives an example item in all four conditions.

To increase the ratio of fillers to experimental items, we added 48 new fillers to those used in the earlier experiments, giving 88 fillers in total. Half of the new fillers were ungrammatical, and 20 of them were followed by a comprehension question.

Table 6: Experimental conditions for the embedded clause stimuli. The Attractor was manipulated for number (SG/PL). The embedded Verb was also manipulated for number to match or mismatch the head noun (always singular), creating Grammatical (SG) and Ungrammatical (PL) conditions. The main verb was always conjugated for the first person singular marking.

Att. Type	Att. Num.	Verb Num.	Sentence & Gloss							
Verbal	**SG**	**SG**	Dün	Tut-tu-u	aç	mutfak-ta	sürekli	zpla-d	diye	duy-du-m
		Grammatical	*yesterday*	*hire*-NMLZ-POSS	*cook*	*kitchen*-LOC	*non.stop*	*jump*-PST	*that*	*hear*-PST-1SG
		PL	*Dün	Tut-tu-u	aç	mutfak-ta	sürekli	zpla-d-lar	diye	duy-du-m
		Ungrammatical	*yesterday*	*hire*-NMLZ-POSS	*cook*	*kitchen*-LOC	*non.stop*	*jump*-PST-PL	*that*	*hear*-PST-1SG
			'Yesterday, I heard that the hired$_{sg}$ cook jumped in the kitchen non-stop.'							
Verbal	**PL**	**SG**	Dün	Tut-tuk-lar-	aç	mutfak-ta	sürekli	zpla-d	diye	duy-du-m
		Grammatical	*yesterday*	*hire*-NMLZ-PL-POSS	*cook*	*kitchen*-LOC	*non.stop*	*jump*-PST	*that*	*hear*-PST-1SG
		PL	*Dün	Tut-tuk-lar-	aç	mutfak-ta	sürekli	zpla-d-lar	diye	duy-du-m
		Ungrammatical	*yesterday*	*hire*-NMLZ-PL-POSS	*cook*	*kitchen*-LOC	*non.stop*	*jump*-PST-PL	*that*	*hear*-PST-1SG
			'Yesterday, I heard that the hired$_{pl}$ cook jumped in the kitchen non-stop.'							

4.3 Procedure

The experiment used the word-by-word moving window self-paced reading method and was run online on the PCIbex platform (Zehr and Schwarz, 2018; Drummond, 2013). Participants were sent a link and had a week to complete the study at their own convenience. They first read and agreed to the consent form, then filled out a brief demographic questionnaire. They were then given instructions and nine practice sentences to familiarize themselves with the task, which consisted of reading sentences at their own pace, giving a yes–no acceptability judgment after each sentence, and answering occasional comprehension questions. Comprehension questions followed a subset of the filler sentences and targeted the content of the sentence just read. Apart from the practice items, no feedback was given on either the acceptability judgments or the comprehension questions. The 40 experimental items were distributed across four lists using a Latin-square design, so that each participant saw one version of each item and ten items in each condition. Participants also saw 88 filler sentences. The experiment took approximately 25 minutes.

4.4 Analysis

Prior to the analyses, we applied the same discrimination criterion as in Experiments 1 and 2, excluding participants whose acceptance rates for grammatical and ungrammatical sentences differed by less than 25 percentage points in the singular-attractor conditions, where the attractor carries no number information. Four participants met this criterion. We additionally excluded two participants whose accuracy on the filler comprehension questions was below 70%. In total, 6 of the 79 participants (7.59%) were excluded, and the final dataset contained 73 participants and 2,920 observations.

All analyses details with respect to 'yes' response were identical to previous experiments. A Bayesian hierarchical model assuming the Bernoulli distribution (logit link) was fit to the binary acceptability response, with the dependent variable coded as 1 for an 'acceptable' rating and 0 otherwise. Grammaticality, Attractor Number, and their interaction were included as fixed effects. The random-effects structure was maximal: by-subject and by-item random intercepts and random slopes for Grammaticality, Attractor Number, and their interaction, including correlations (Barr et al., 2013). Grammaticality and Attractor Number were sum-coded at ±0.5, as in Experiments 1 and 2, so that each slope is the full difference between the two levels of a factor. The priors were identical to those of the acceptability models in Experiments 1 and 2: $\mathcal{N}(0, 1)$ on the fixed effects, $\mathcal{N}(0.85, 0.70)$ on the intercept, Exponential(1) on the random-effect standard deviations, and LKJ(2) on the random-effect correlations. The model was estimated with four chains of 12,000 iterations (2,000 warmup).

Reading times were log-transformed prior to analysis. A separate Bayesian hierarchical model assuming the Gaussian distribution was fit to log reading time at each sentence region, using the same fixed-effects specification (Grammaticality, Attractor Number, and their interaction) and the same maximal by-subject and by-item random-effects structure. Predictors were sum-coded. Inference focused on the critical verb and the spillover region. The priors, sampler settings, and full posterior summaries for every region are reported in Appendix B.

4.5 Results

4.5.1 Acceptability judgments

Participants judged the filler sentences accurately, accepting 95.9% of grammatical fillers (CI = [95.2, 96.6]) and rejecting 86.6% of ungrammatical fillers (CI = [85.4, 87.8]). They also answered the comprehension questions accurately (M = 91.0%, CI = [89.4, 92.4]), indicating that participants read the sentences attentively.

Figure 4A shows the mean proportion of 'yes' (acceptable) responses by condition, with 95% Clopper–Pearson confidence intervals. The pattern replicates the verbal-attractor results of Experiments 1 and 2. Grammatical sentences were accepted at comparable rates regardless of attractor number (plural: M = 89.3%, CI [86.8, 91.5]; singular: M = 89.0%, CI [86.5, 91.2]). The same insensitivity to attractor number held in the ungrammatical conditions, where attraction would surface as a increased acceptability in plural-attractor conditions: ungrammatical sentences were accepted no more often with plural attractors (M = 13.6%, CI [11.2, 16.3]) than with singular attractors (M = 12.9%, CI [10.5, 15.5]).

Overall error rates were higher in Experiment 3 than in the speeded acceptability experiments (12.0%, CI [10.9, 13.3], versus 5.4% CI [4.5, 6.6] in the verbal conditions of Experiment 1 and 6.0% CI [5.2, 6.9] in Experiment 2). This increase was spread across all four conditions rather than driven by any single condition. The same holds when we consider only the baseline singular-attractor conditions, where the attractor carries no number information and errors therefore index baseline task difficulty alone (11.9%, CI [10.3, 13.7], against 5.5% CI [4.1, 7.1] and 6.9% CI [5.7, 8.3] in Experiments 1 and 2). Error rates of similar magnitude occur in the singular conditions of Türk and Logačev (2024) (8.9%, CI [7.8, 10.2]), and nominal attractors nonetheless produced an attraction effect of 10.6 percentage points on top of that baseline. Thus, we conclude that overall increase in errors would not be likely to hide otherwise present attraction. If anything, the higher baseline provides more room for false alarms to increase with plural attractors (Hammerly et al., 2019; Ratcliff, 1978).

These descriptive results were confirmed by the Bayesian model fitted to ‘yes’ responses, whose fixed-effect posteriors are shown in Figure 4B. The model showed strong evidence for a main effect of grammaticality ($\beta = 4.89$, 95% CrI $[4.32, 5.46]$, $P(\beta<0) > 0.001$) and, critically, no evidence for the grammaticality by attractor number interaction that would index attraction ($\beta = 0.06$, 95% CrI $[-0.50, 0.62]$, $P(\beta<0) = .42$). A Bayes factor computed by bridge sampling, comparing this model against one omitting the interaction, favored the null ($BF_{01} = 3.11$).

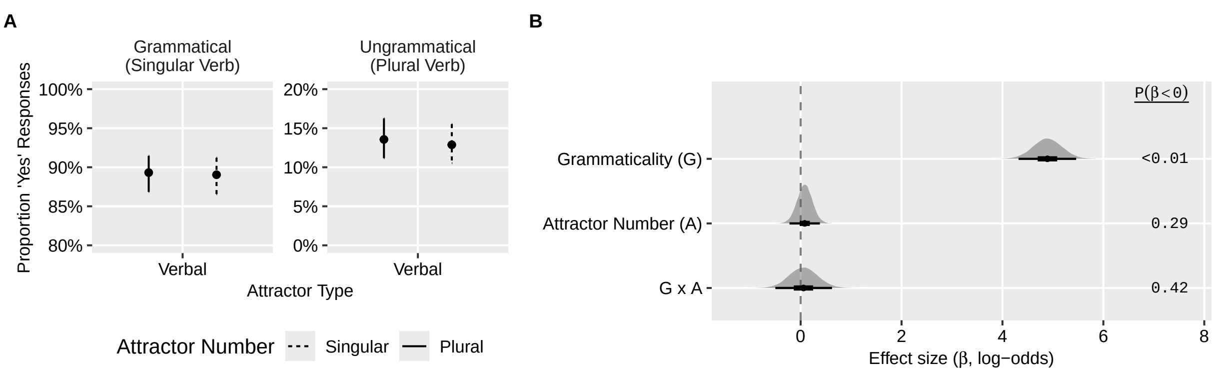


Figure 4: Experiment 3 acceptability results. A: Mean proportion of ‘acceptable’ responses by grammaticality and attractor number. Error bars show 95% Clopper–Pearson confidence intervals. B: Posterior means and 95% credible intervals for the fixed effects of the acceptability model, on the log-odds scale; the right column gives the posterior probability that each coefficient is negative.

4.5.2 Reading times

Figure 5 presents region-by-region reading times by condition, with the critical verb and the spillover region additionally shown enlarged. In the text below, descriptive reading times are untransformed condition means in milliseconds with standard errors following Morey and Cousineau (Morey, 2008; Cousineau, 2005). Model estimates ($\hat{\beta}$) are on the log reading-time scale and are summarized by posterior means with 95% credible intervals (CrI).

Reading times in the pre-critical regions showed the expected cost of plural morphology. At the attractor region, plural attractors were read more slowly than singular ones (plural mean = 650 ms, SE = 12.5; singular mean = 569 ms, SE = 9.1; $\hat{\beta} = 0.090$, CrI = $[0.054, 0.125]$, $P(\beta>0) > .999$). Since this difference precedes the critical verb, we take it to reflect the additional processing cost of reading plural marking rather than any agreement process, as has been observed previously (Wagers et al., 2009). This cost did not persist into the pre-verbal regions, where the numerical difference reversed in the first adjunct after the head noun ($\hat{\beta} = -0.021$, CrI $= [-0.047, 0.004]$, $P(\beta{>}0) = .05$). Moreover, we did not find any evidence for an effect of attractor number in the head noun region ($\hat{\beta} = 0.012$, CrI $= [-0.018, 0.042]$, $P(\beta{>}0) = .79$). Two pre-critical regions, the attractor and the head, additionally showed a small difference by grammaticality, with grammatical conditions read more slowly (attractor region: $\hat{\beta} = 0.038$, CrI $= [0.005, 0.071]$, $P(\beta{>}0) = .988$; head region: $\hat{\beta} = 0.036$, CrI $= [0.005, 0.066]$, $P(\beta{>}0) = .988$). As the conditions are string-identical up to the critical verb, we attribute these to residual variance. Posterior summaries for these and all other non-critical positions are reported in the appendix.

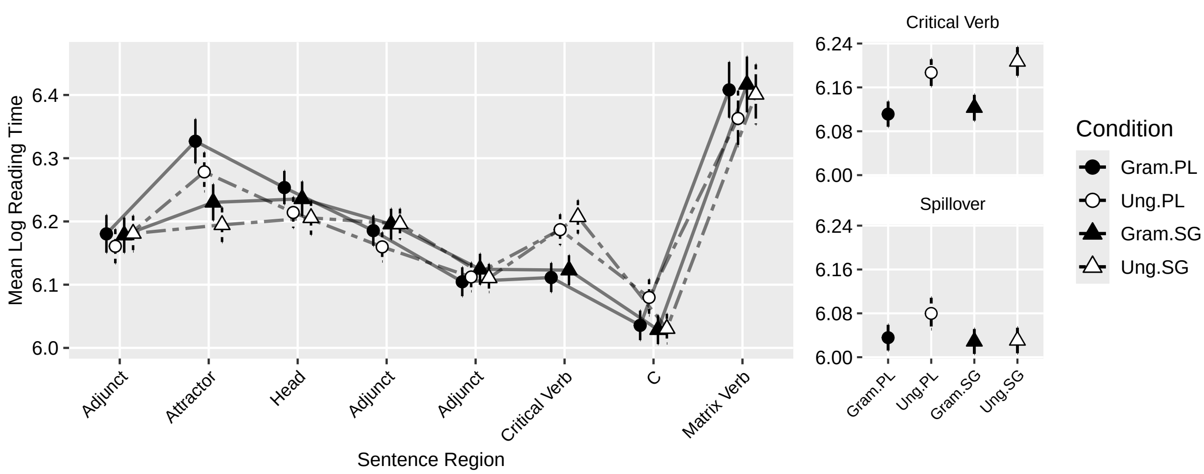


Figure 5: Experiment 3 reading times by region and condition (log scale), with the critical verb and spillover regions enlarged. Error bars show within-subject 95% confidence intervals (Hope, 2022).

Posterior estimates for the two regions of interest are shown in Figure 6, which presents the coefficients for grammaticality, attractor number, and their interaction at the critical verb and the spillover region. At the critical verb, the model provides strong evidence for a main effect of grammaticality: ungrammatical sentences were read more slowly than grammatical ones (ungrammatical mean = 555 ms, SE = 8.9; grammatical mean = 505 ms, SE = 8.2; $\hat{\beta} = -0.083$, CrI $= [-0.113, -0.053]$, $P(\beta{<}0) > .999$), confirming that participants registered the agreement violation. Critically, there was no substantial evidence for a main effect of attractor number at the verb ($\hat{\beta} = -0.017$, CrI = $[-0.047, 0.014]$, $P(\beta{<}0) = .866$), and no evidence for the grammaticality by attractor number interaction that would index attraction ($\hat{\beta} = 0.012$, CrI = $[-0.042, 0.067]$, $P(\beta{>}0) = .676$): the small numerical plural-attractor speedup was of comparable magnitude in ungrammatical ($\Delta \log = -0.020$) and grammatical ($\Delta \log = -0.012$) sentences. A Bayes factor comparing this model against one omitting the interaction gave very strong support for the null ($BF_{01} = 36.27$).

In the spillover region, the grammaticality effect attenuated and, however, our model still exhibited very weak evidence for the main effect ($\hat{\beta} = -0.022$, CrI = $[-0.050, 0.006]$, $P(\beta{<}0) = .941$). The model suggests strong evidence for a main effect of attractor number, with plural attractors overall being read more slowly ($\hat{\beta} = 0.028$, CrI = $[0.002, 0.054]$, $P(\beta{>}0) = .984$), and a weak interaction whose posterior mass falls in the direction opposite to facilitation ($\hat{\beta} = -0.043$, CrI = $[-0.099, 0.013]$, $P(\beta{<}0) = .935$). Descriptively, the plural-attractor slowdown was confined to ungrammatical sentences (singular mean = 446 ms, SE = 7.6; plural mean = 509 ms, SE = 23.2) and absent in grammatical ones (singular mean = 452 ms, SE = 10.5; plural mean = 456 ms, SE = 10.7). A Bayes factor again favored the model without the interaction ($BF_{01} = 12.26$). Given the small posterior mass involved and the dispersion in the ungrammatical plural cell, we do not interpret this as evidence for an inhibitory effect of the attractor. We note, however, that a numerically similar reversal, i.e. slower spillover reading times with plural attractors in ungrammatical sentences, has been reported by Xu and Futrell (2025), who attribute it to reanalysis.[3] They argue that comprehenders notice the misprocessing at the verb and pay the cost one region downstream. In their data the reversal followed attraction at the verb itself; in ours there is no attraction at the verb, so a reanalysis account would have to take a different form: upon detecting the agreement violation, readers may re-examine the plural-marked attractor in their working-memory as a potential licenser and fail, for verbal *-lAr* without an overt head noun following marginally license plural agreement, and the syntactic configuration our materials are presented in prohibits this ability al-

[3]Similar reversal effects were also visited other dependency resolution (Cunnings and Felser, 2013; Sturt, 2003) as well as agreement attraction studies (Wagers et al., 2009), see Jäger et al. (2017) for a detailed meta-analysis.

together. On this reading, the plural surface form is visible to deliberate reanalysis even though it does not enter into initial dependency formation. Given the weak posterior support, we offer this only as a speculation to be tested in designs with more power at the spillover region.

No credible effects were observed at the matrix verb, where all posterior probabilities fell below .79 and credible intervals were wide throughout; full estimates are given in the appendix.

Taken together, the reading time results converge with the acceptability results. The grammaticality effect at the critical verb shows that the violation was detected, but at neither the verb nor the spillover region did the plural attractor facilitate the processing of ungrammatical sentences, confirming the absence of attraction with verbal attractors.

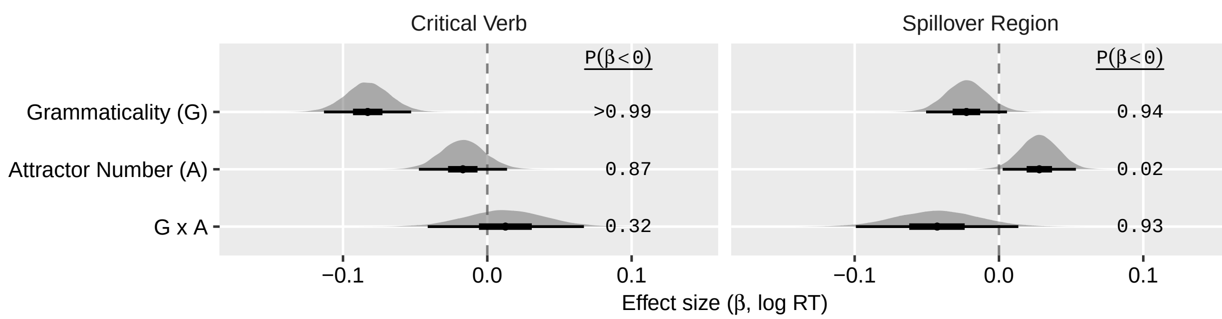


Figure 6: Posterior distributions of the fixed effects of the log reading-time models at the critical verb and the spillover region. Slabs show posterior densities with 50% and 95% intervals; the right column gives the posterior probability that each coefficient is negative.

5 General Discussion

The goal of the present study was to investigate how speakers resolve agreement dependencies and what kinds of information they rely on in doing so, using erroneous decisions in speeded acceptability judgment tasks and reading-time profiles in self-paced reading as diagnostics. A better understanding of the mechanisms that give rise to these errors can inform theories of memory retrieval during sentence processing and, more broadly, the architecture of the human sentence processing system.

More specifically, we asked whether surface-form overlap alone can trigger agreement attraction during comprehension. Turkish provides a strong test case because the plural morpheme *-lAr* appears on both nouns and nominalized verbs. Three properties of the attractors must be kept apart here: being contained within the subject DP, being able to occur as a subject in some construction of the language, and being able to control agreement. In our materials, neither attractor type is itself the matrix subject; both are embedded inside the subject DP as modifiers of the head noun. Both types can also occur as subjects elsewhere: nominalized verbs as headless reduced relative clauses, and genitive nominals as subjects of embedded possessive clauses. The two diverge only in the third property: genitive nominals control agreement in the environments where they act as subjects, whereas strings bearing verbal *-lAr* do so only marginally, if at all. If surface-form overlap alone is sufficient to trigger attraction, then we should observe attraction from both nominal and verbal attractors. We would also expect comparable attraction effects in both conditions if attraction were driven by similarity to a possible subject. If, however, attraction is gated by abstract controller features, then we should observe attraction only from nominal attractors, which can be controllers, and not from verbal attractors, which cannot. The results strongly supported the latter prediction: robust attraction was observed for nominal attractors but not for verbal attractors, despite maximal surface similarity.

Across two speeded acceptability judgment experiments and a self-paced reading experiment, nominalized verbal attractors did not induce agreement attraction. In Experiment 1, we observed robust attraction for nominal attractors, closely replicating previous nominal findings (Lago et al., 2019; Türk and Logačev, 2024), but found no attraction for verbal attractors. Importantly, this was not merely a smaller-than-nominal effect: follow-up analyses restricted to the verbal conditions, together with the Bayes factor analysis, provided strong evidence for the absence of verbal attraction. In Experiment 2, which removed the nominal conditions and retained only the verbal ones, the null pattern replicated. Ungrammatical sentences with plural nominalized verbal attractors were accepted no more often than their singular-attractor controls. Finally, in Experiment 3, we tracked the same verbal manipulation word-by-word in self-paced reading and found no reading-time signature of attraction: at the critical verb, ungrammatical sentences were read more slowly regardless of attractor number, with no evidence for the attraction interaction; at the spillover region, the only hint of an attractor effect was a weak, non-credible slowdown for plural attractors in ungrammatical sentences—the direction opposite to attraction-driven facilitation.

Verbal attractors share the surface phonology of plural nouns and can stand as subjects, yet they produced no attraction. This points to a division of labor. Phonological overlap demonstrably affects general reading and memory processes: when the subjects of two clauses are phonologically similar, readers slow down and comprehension suffers (Acheson and MacDonald, 2011). But the same kind of overlap does not appear to enter into the resolution of syntactic dependencies: holding phonologically similar words in memory does not disrupt the completion of a filler-gap dependency (Kush et al., 2015), and in our data, sharing a phonological form with agreement controllers did not disrupt—or illusorily license—the computation of subject-verb agreement, contrary to the proposal advanced by Slioussar (2018). The pattern converges with findings from English and Czech showing little to no attraction from pseudo-plural or phonologically plural forms (Bock and Eberhard, 1993; Lacina and Chromỳ, 2022), but it appears to diverge from findings in Russian (Slioussar, 2018). We propose that the critical factor is not surface overlap per se, but whether the attractor form is associated with being a possible agreement controller (Lago et al., 2019; Bhatia and Dillon, 2022; Bleotu and Dillon, 2024). Russian genitive nouns, despite their surface form, can control agreement in other constructions (Partee and Borschev, 2004), whereas our Turkish verbal attractors and Czech genitive nouns cannot. On this view, controller compatibility, rather than phonological overlap alone, determines whether attraction arises.

In the remainder of the General Discussion, we consider how our results, together with previous findings on case syncretism and phonological overlap, bear on different theoretical accounts of agreement attraction. We then address limitations of the present study and outline directions for future research.

5.1 Accounts of agreement attraction and the role of surface-form overlap

In deciding which theoretical accounts to evaluate against our findings, we follow the large-scale model comparison of Yadav et al. (2023), who implemented seven computational models of number agreement: three variants of representation distortion models, two cue-based retrieval models, and two hybrids models. They evaluated them against 17 benchmark reading-time datasets from four languages. The best quantitative fit was achieved by a hybrid model combining feature percolation with cue-based retrieval; the second-best fit was achieved by a Marking and Morphing variant augmented with a grammaticality bias, operationalizing the proposal of Hammerly et al. (2019). We therefore concentrate on these two model classes. We first discuss the limitations of the Marking and Morphing account with respect to surface-form effects, and then turn to cue-based retrieval theory. Because the hybrid model of Yadav et al. (2023) inherits its percolation component from Marking and Morphing, we restrict our discussion to its retrieval component.

5.1.1 Marking and Morphing Account

The Marking and Morphing model, as outlined in the Introduction, computes agreement from a reconciled number value of the subject phrase. This model can predict that a plural attractor should induce some attraction regardless of its case marking, but it does not predict the *additional* boost previously observed for syncretic attractors in cases like German or Czech (Hartsuiker et al., 2003; Lacina et al., 2025). Importantly, Eberhard et al. (2005) distinguish among three sources of number information: grammatical marking (whether a noun requires plural agreement on verb), morphological number (the presence of plural-like morphology on inherently singular forms such as *scissors*), and notional number (the conceptual plurality of collective nouns such as *gang* or *fleet*). For the head noun, all three sources contribute to the reconciled number value. For the attractor, however, only grammatical marking—the overt plural morpheme—is relevant; morphological and notional number of the attractor do not participate in the morphing process. The spreading mechanism is thus unaffected by whether the attractor's form resembles a nominative; the only relevant property is whether the attractor carries a marked number feature. It is difficult to reconcile the previous syncretism-driven effects with the model's exclusive reliance on grammatical number marking, and to our knowledge, no published extension of the account has attempted to do so (Avetisyan et al., 2020). The grammaticality-bias variant of Hammerly et al. (2019) inherits the same limitation. In their model, Hammerly et al. (2019) introduced an a priori bias parameter shifts the mapping from the reconciled number value to responses through evidence accumulation following Ratcliff's theory of Drift Diffusion. However, it leaves the number sources themselves untouched, so case syncretism still has no channel through which to operate.

For our verbal attractor manipulation, the predictions of this model family depend on how the parser encodes *-lAr* on the nominalized verb. In our items, the nominalized verb always appears with an overt singular head, so *-lAr* can only realize verbal agreement rather than nominal number.[4] The attractor therefore carries no grammatical nominal plural marking, there is nothing overt to percolate, and the model predicts the null result we observed. The same reasoning carries over to the percolation component of the hybrid model of Yadav et al. (2023): with no nominal plural feature on the overt attractor, the distortion pathway is inert, and any attraction would have to arise in the retrieval component, to which we now turn.[5]

5.1.2 Cue-Based Retrieval Accounts

Cue-based retrieval models provide the most explicit mechanistic account of previous effects of case syncretism on agreement attraction in comprehension (Lewis and Vasishth, 2005; Smith and Vasishth, 2020; Engelmann et al., 2019). One natural approach to accounting for case syncretic effects within a cue-based retrieval model is to assume that the retrieval cue for case is matched against the morphological form of the noun phrase rather than its abstract case feature (cf. Slioussar, 2018). This assumption goes beyond the canonical model of Lewis and Vasishth (2005), in which retrieval cues are a subset of the abstract morphosyntactic features encoded on chunks—

[4] When the head is dropped, verbal *-lAr* can indicate the plurality of an entity and can license reciprocals such as *birbirleri* 'each other'. This configuration is not used within our materials.

[5] The relative clause does contain a pro-dropped subject that must be plural in order to license plural verbal agreement. Whether pro-dropped plural subjects can themselves act as attractors is, to our knowledge, an open question, and we leave it to future work.

case, number, syntactic category—and which does not address case syncretism or the morphological form of case exponents. If the attractor's surface case is syncretic with the nominative—as in Russian and German, where accusative and nominative forms of certain genders are identical—then the attractor matches the [+NOM] retrieval cue in addition to [+PL], producing stronger interference than a distinctively marked attractor. This aligns with findings in German (Hartsuiker et al., 2003), Czech (Lacina et al., 2025), and Russian (Slioussar, 2018), where syncretic attractors consistently produce more attraction than non-syncretic ones.

Slioussar (2018) proposed a more radical mechanism in which phonological overlap itself can activate competing feature bundles. On her account, a Russian genitive singular noun whose surface form is homophonous with the nominative plural activates not just the chunk corresponding to [+GEN, +SG] but also a competitor chunk carrying [+NOM, +PL]. This co-activated competitor then enters the retrieval race alongside the true target and the veridical encoding of the attractor, increasing the probability of a misretrieval of this co-activated competitor at the verb. The mechanism is grounded in well-established principles of spoken word recognition—cohort competition, phonological neighborhood effects—and generalizes naturally from lexical access to syntactic retrieval.

This proposal predicts that any accidental phonological overlap with a controller-relevant form should always increase attraction, including the overlap between verbal and nominal *-lAr* in Turkish tested here.[6] However, the results of the current study failed to support this prediction: attraction was entirely absent when the attractor was verbal marked with plural *-lAr*. These results are consistent with the results reviewed above from Czech, which exhibits the same genitive singular / nominative plural homophony as Russian, and yet Lacina and Chromý (2022) found no attraction from the overlapping forms, and later work by Lacina et al. (2025) confirmed that only case-syncretic—not merely phonologically overlapping—attractors produce attraction. Prior work on argument structure showing that phonological similarity between verb forms does not modulate

[6]Since the overlap between verbal and nominal *-lAr* is not purely accidental homophony and a genuine identity of form produced by the same phonological string for very similar functional meanings, we expect this particular experiment in Turkish to be particularly robust.

thematic processing (Kush et al., 2015), also provides evidence that the parser does not routinely exploit phonological overlap during structure building.

The same dissociation between abstract morphosyntactic similarity and phonological similarity emerges from the literature on similarity-based interference in center-embeddings, introduced in §1.2. Uehara and Bradley (1996) found that Japanese multiply center-embedded sentences become harder to process as the number of nominative *-ga*-marked subjects increases. In another experiment using Korean, repeating phonologically identical case markers on noun phrases whose syntactic roles differed produced no comparable difficulty, leading them to attribute the Japanese effect to syntactic rather than phonological similarity (see also Uehara 2003). Fedorenko et al. (2004) unconfounded the two factors in Russian, where surface case marking and abstract case can be crossed: interference arose only when two noun phrases matched in both morphological marking and abstract case, whereas identical surface marking alone had no effect. Work on case attraction in German points the same way, implicating abstract case rather than its surface exponent (Bader et al., 2000). Logačev and Vasishth (2012) showed experimentally that case-matching effects are driven by abstract case proper rather than by surface properties canonically associated with it. Most directly relevant here, in Turkish, Bakay (2020) independently manipulated syntactic and phonological case interference using center-embeddings and found in eye-tracking that reducing the number of noun phrases sharing abstract case facilitated processing, whereas the phonological overlap of the case markers had no effect. Across dependency types and languages, then, what generates interference is shared morphosyntactic specification, not shared phonological form. Indeed, in canonical cue-based retrieval models, it is morphosyntactic and semantic features that guide memory access (Lewis and Vasishth, 2005; Van Dyke and Lewis, 2003); and phonological form influences retrieval only insofar as it determines which morphosyntactic features are assigned to the chunk.

A more parsimonious alternative to explain case syncretism effects, introduced in §1.3, is that what the parser tracks is whether the surface form of the attractor is associated with being a possible agreement controller in that language (Lago et al., 2019). On this view, the relevant quantity is not phonological similarity but the distributional statistics of the form—how often the morphological shape of the attractor appears on agreement controllers. Lago et al. (2019) argued that the reason Turkish genitive marking induces attraction while English genitive marking does not is that Turkish genitives, but not English ones, regularly function as subjects in embedded clauses.[7] In our experiments, the verbal attractors carry no comparable association. Configurations in which a reduced relative clause controls agreement on the matrix verb are rare and independently marked, so the surface form of our attractors offers the parser little evidence of controllerhood. A controller-association account therefore predicts precisely the null effect we observe. Other Turkish findings fit this picture as well: Ulusoy (2023) observed attraction from plural matrix-clause subjects into embedded verbal agreement—attractors which, as matrix subjects, are themselves canonical agreement controllers. However, extra-clausal attractors that are not marked with a case marking that can control agreement do not induce attraction effects neither in reading times nor accuracy. Controllerhood account also provides a natural account of the cross-linguistic variation between the pattern of agreement attraction in Russian and Czech. As discussed in §1.3, Russian genitives retain both controller and subjecthood associations that Czech genitives lack. This explains why the same homophony can modulate attraction in one language but not the other.

Findings from other languages reinforce the language-general, distributional tracking component of such an account. In a series of six experiments, Schlueter et al. (2018) showed that the English conjunction *and*, which they argue to be a reliable statistical correlate of plurality, increases acceptance of ungrammatical plural verbs even in the absence of any overt plural morphology, including when it merely conjoins two adjectives modifying a singular head noun (*the slogan about the loyal and caring husband*). Such effects do not bear on controllerhood directly, but they show that retrieval cues need not target categorical features of the controller: features that are merely statistically associated with the cue suffice to activate an item in memory. The

[7] English genitive pronouns can of course serve as subjects of embedded clauses (*His arrival was delayed*). The critical difference is that English genitive pronouns do not trigger a morphological reflex on the embedded verb, in other words English lacks the possessive agreement paradigm that Turkish has. Therefore, the surface association between genitive marking and controllerhood is far weaker. Lago et al.'s (2019) account can then be stated more precisely in terms of controller association rather than mere subjecthood following Türk and Logačev (2024).

Romanian findings of Bleotu and Dillon (2024) make the same point with a subjecthood-related cue. Romanian generally prohibits bare nouns from surfacing as preverbal subjects, so a bare-noun attractor is distributionally less subject-like than a determiner-bearing one; correspondingly, full DP attractors produced more attraction than bare nouns. Bleotu and Dillon (2024) conclude that the cues guiding retrieval at the verb are language-specific, learned from the distribution of forms in subject position. On this broader view, comprehenders tune the weights of retrieval cues to their linguistic experience, and the precise cues that matter vary from language to language (Dillon and Keshev, 2024; Engelmann et al., 2019).

The Hindi data from Bhatia and Dillon (2022) sharpen this picture in a different way, by showing that a language-general form statistic can be necessary without being sufficient. Agreement in Hindi is relational: the verb agrees with the structurally most prominent case-unmarked argument, so that, depending on the case frame, the controller may be the subject or the object (Bhatia, 2019; Bhatt, 2005; Mahajan, 1990). Controllerhood in Hindi is therefore not tied to a grammatical role, but it is tied to a surface property—the absence of overt case marking—and comprehenders are demonstrably sensitive to that property. In Experiments 1 and 2, Bhatia and Dillon (2022) found attraction from bare plural attractors in both embedded subject and embedded object positions, precisely where the language-wide association between bareness and controllerhood licenses them as candidates.

Bareness alone, however, did not suffice. In Experiment 3, an attractor bearing overt *-ko* induced no attraction, as expected if the form-level licence is absent; but a bare attractor that was not itself controlling agreement in its clause induced no attraction either, even though the same form had done so in Experiments 1 and 2, where it was the local controller. Experiment 4 makes the same point within a single sentence: an embedded subject that controlled agreement in its own clause induced attraction, whereas a matrix object did not, because the matrix subject was already controlling agreement there. What interferes is thus an item that is both formally eligible and locally realizing controllerhood, and Bhatia and Dillon propose that Hindi speakers proactively encode a [+CONTROLLER] feature on such items as the sentence unfolds.

These findings point to a bifurcation between language types in the scope over which controller association is evaluated, and German may pattern with Hindi in this respect. Hartsuiker et al. (2003) found that accusative-marked plural attractors induced extra attraction beyond the base plural effect, but only for feminine nouns, where the accusative determiner *die* is syncretic with the nominative; in masculine and neuter conditions no syncretism-driven boost was observed. A language-general association between *die* and controllerhood would predict a boost across the board; the gender-specificity instead suggests that the attractor's form was evaluated against the controller form relevant in that particular sentence.

We therefore expect the scope of the controller association to vary across languages. In some, as in Hindi and possibly German, it is narrow, and the attractor's status as a controller is evaluated within the sentence being processed. In others, as in Turkish, it is not restricted to the sentence at hand: genitive marking regularly appears on agreement controllers in the language, and verbal *-lAr* never does, so the distribution of the form across the language is itself sufficient to determine what the parser tracks (Dillon and Keshev, 2024; Engelmann et al., 2019). What our data rule out is the claim that surface-form overlap alone—divorced from both controllerhood and syntactic licensing—is sufficient to generate agreement attraction.

How might this controller-sensitivity be implemented within cue-based retrieval? A concrete possibility is that the retrieval probe at the verb includes a controller cue alongside its number and case cues, and that a chunk's match to this cue is graded rather than categorical: the degree of match reflects the strength of association between the chunk's morphological form and controllerhood, whether that association is estimated from the distribution of the form in the language or from its behavior in the current sentence. Graded cue–feature match of this kind is independently motivated within the activation-based framework of Engelmann et al. (2019), where match strength is continuous and modulated by experience. On this implementation, a Turkish genitive attractor partially matches the controller cue—genitive forms regularly control embedded agreement—in addition to matching the number cue, so interference arises. A verbal *-lAr* attractor matches the number cue in form only, and its controller association is effectively zero, so no attraction is predicted;

the Russian–Czech contrast follows in the same way from the differing controller associations of genitive marking in the two languages. This sketch leaves open how controller associations are acquired and parameterized; a fully explicit implementation would need to derive the association strengths from corpus distributions, which we leave to future modeling work.

5.2 Limitations and alternative explanations

Although our results align with the previous cross-linguistic findings and converge on a more parsimonious account of surface-overlap compared to phonological modulation, certain reservations are in order in view of the specific linguistic constructions we are using, the presence of ceiling effects, and the nature of the null effects.

5.2.1 Nominal vs. nominalized verb distinction

Our attractors included genitive modifiers (e.g., *milyonerlerin*, 'millionaires') and reduced relative clauses (e.g., *gördükleri*, 'the one(s) they saw'). If surface-form overlap alone were sufficient to trigger attraction, both types should produce attraction, since they share the same plural morpheme *-lAr* and can both surface as subjects. However, the two types of attractors differ in their linguistic status in ways that deserve attention. We believe that although this distinction is an important aspect to control in future experiments, it does not undermine the main conclusion of the paper.

The initial issue concerns the categorial identity of the verbal attractors. Our verbal attractors are nominalized forms: they take case marking, can be modified by adjectival elements, and occupy argument positions—all properties parallel to noun phrases (Göksel and Kerslake, 2005). However, they are also propositional in the sense that their syntactic and semantic properties are derived from their verbal source (Satik, 2024; Demirok, 2019; Lees, 1965; Kornfilt, 1984, 2003; Kornfilt and Whitman, 2011). Demirok (2019) argues that nominalized *-DIK* phrases denote propositions, whereas *-MA* phrases denote events, and only the latter can freely serve as matrix subjects. Importantly, however, this propositional characterization does not carry over to the *-DIK* forms used in our experiment: reduced relative clauses that modify a head noun are proposed to denote entities rather than propositions (Demirok, 2019). The strongest evidence comes from coordination with *ile* ('and/with'), which in Turkish can only coordinate two noun phrases. As shown in (12), propositional *-DIK* phrases cannot be coordinated with *ile* (12a), but *-DIK* phrases built on a (dropped) head noun—parallel to our verbal attractors—can be (12b), suggesting that in this use they pattern syntactically as noun phrases.

(12) a. * [Suzan'n hata-y bul-du-u] ile [Merve'nin email-i yaz-d-n-] biliyorum.

* [Suzan'n	hata-y	bul-du-u]	ile	[Merve'nin	email-i	yaz-d-n-]	biliyorum.
Susan-GEN	mistake-ACC	find-DIK-3SG.POSS	with	Merve-GEN	email-ACC	write-DIK-3SG.POSS-ACC	I.know

Intended: 'I know that Susan found the mistake and Merve wrote the email.'

b.

[Sen-in	piir-di-in]	ile	[ben-im	al-d-m]	masa-da.
you-GEN	cook-DIK-2SG.POSS	with	I-GEN	buy-DIK-1SG.POSS	table-LOC

'The thing you cooked and the thing I bought are on the table.'

In sum, the contrast between our nominal and verbal attractors cannot be reduced to a simple verb-versus-noun confound. Both our nominal and verbal attractors occupy nominal positions and bear nominal syntax. The *-DIK* forms used in our experiment are entity-denoting relative clauses—the most noun-like end of the nominalization spectrum. If the parser's sensitivity to category were the sole driver of our null result, it would have to operate at a finer grain than the noun/verb boundary, distinguishing among subtypes of nominal expressions based on their derivational history. This remains possible, but controllerhood is the more parsimonious account, since it explains the same contrast without positing category sensitivity at that grain.

Even granting this theoretical distinction, the processing evidence does not establish that derived and simple nouns must behave differently. One might argue that the parser carries the verb/noun distinction into derived structures, and that our null effect therefore reflects category sensitivity rather than controllerhood. Some evidence is consistent with this view: Momma et al. (2020) show that deverbal nouns do not compete with simple verbs in a sentence-picture interference paradigm, regardless of semantic similarity, suggesting sensitivity to final categorization (see also Dell et al., 2008). Other work points in the opposite direction: Tsaprouni and Manouilidou (2025) find that the parser ignores a derived word's derivational history unless its eventive properties are carried over (see also Stockall and Manouilidou, 2014), and Gaston et al. (2020) show that category-mismatching lexical competitors are activated rather than suppressed in category-constraining contexts, indicating that category constraints facilitate correct-category candidates rather than inhibit wrong-category ones. In short, the processing literature does not converge: sometimes derivational history matters, sometimes it does not, and the conditions governing this variation remain poorly understood.

Moreover, regardless of where one lands on the category question, the surface-form properties of our verbal attractors should have provided maximal conditions for phonological modulation. The *-lAr* suffix on reduced relative clause verbs is string-ambiguous for number: although syntactically it marks agreement with an implicit external plural argument, the surface string does not disambiguate the two functions. A parser encountering *gördükleri* ('the one(s) they saw') cannot determine from the string alone whether *-lAr* encodes nominal plural on the entity that was seen or the verbal agreement that the subject of the relative clause is plural, or both. This ambiguity is precisely the kind of surface condition that a phonological co-activation account should exploit—the form *-lAr* is the same morpheme, in the same string position, on a categorially nominal expression. The broader phonological interference literature supports the expectation that such overlap should produce confusion, whether through transient activation of competing representations in continuous speech processing (McClelland and Elman, 1986; Norris, 1994) or through phonological similarity interference in working memory (Baddeley, 1966; Conrad, 1964; Acheson and MacDonald, 2009). Despite all of this, attraction is absent, setting a strong upper bound on what surface overlap alone can contribute.

5.2.2 Decreased grammaticality with singular verbal attractors

One unexpected finding in Experiment 2 was that grammatical sentences with singular verbal attractors (13b) were accepted less often than their plural-attractor counterparts (13a). Under any account of attraction, we would expect the opposite pattern: plural attractors should, if anything, reduce acceptability in grammatical conditions by inducing ungrammaticality illusions. We consider two possible explanations for the observed pattern: (i) the singular attractor conditions may have been slightly less natural due to referential properties of singular versus plural reduced relative clauses, and (ii) a possible correlative parse ambiguity in the singular attractor conditions.

(13) a. Tut-tuk-lar-ı aşçı mutfak-ta sürekli zıpla-dı.
hire-NMLZ-PL-POSS cook kitchen-LOC non.stop jump-PST
'The cook that they hired jumped in the kitchen non-stop.'

b. Tut-tu-u aşçı mutfak-ta sürekli zıpla-dı.
hire-NMLZ-POSS cook kitchen-LOC non.stop jump-PST
'The cook that (s/he) hired jumped in the kitchen non-stop.'

The first possibility is that the singular attractor conditions were slightly less natural overall. Singular reduced relative clauses like *tuttuu* ('the one [someone] hired') presuppose a unique referent, whereas plural reduced relative clauses like *tuttukları* ('the one(s) that they hired') can be interpreted with an arbitrary subject, often characterized as PRO_{arb}, and are thus referentially less restrictive (Chomsky, 1993; Göksel, 2005). Plural agreement in Turkish can yield a generic or impersonal reading, effectively downgrading the specificity of the subject referent (e.g., *Burada çok çikolata yerler*, 'They eat a lot of chocolate here', where the third-person plural verb conveys an arbitrary, non-referential agent). This property is not unique to Turkish: in Italian, for instance, subjectless third-person plural forms can similarly express a generic interpretation (Cinque, 1988; Mantovan, 2022), as in (14).[8]

(14) Qui, *pro* lavor-ano anche di sabato.
here *pro* work-3PL even of Saturday
'Here they work even on Saturday.' (adapted from Cinque 1988: 545)

This difference may have been amplified by the fact that nearly all of our subject head nouns referred to occupations: participants may have interpreted the plural *-lAr* marking as referring to a non-referential group of people associated with the profession, making the plural attractor conditions pragmatically more felicitous.

[8] We thank Furkan Dikmen for drawing our attention to this parallel.

A second possibility is that *-DIK*-marked reduced relative clauses can receive a correlative parse, as illustrated in (15), taken from Demirok (2017). If participants initially parsed the *-DIK* phrase as a correlative and assigned a 'who(ever) hired' reading, this could have introduced some notional plurality into the singular attractor conditions, producing a small but systematic increase in 'unacceptable' responses. However, this explanation is unlikely, because plural-marked reduced relative clauses can equally receive a correlative parse: if correlative ambiguity were responsible for the observed pattern, it should have appeared in the plural attractor conditions as well, which it did not.

(15) Mary [John-un piir-di-in-i] ye-di.
Mary John-GEN cook-NMLZ-3SG-ACC eat-PST
'Mary ate what John cooked.'

In either case, the key point is that the observed pattern does not undermine our main conclusion about the absence of attraction from verbal *-lAr*. The singular attractor conditions were still accepted at a very high rate (over 90%), and the critical comparison between singular and plural attractors within the ungrammatical conditions shows no evidence of attraction. The unexpected pattern in the grammatical conditions may reflect subtle pragmatic or parsing factors, but it does not affect the interpretation of the null effect for verbal attraction.

5.2.3 The absence of pseudoplural attraction in English

An important starting point for our study was the observation, reviewed in §1.2, that English pseudoplurals—singular nouns such as *cruise* whose phonological form coincidentally resembles a plural—do not induce agreement attraction (Bock and Eberhard, 1993).

The absence of pseudoplural attraction requires comment under the account proposed here: if attraction is modulated by the association between a surface form and controllerhood, attractors whose form coincides with that of a plural might be expected to interfere. The materials in Bock and Eberhard (1993), however, do not provide a direct test of this prediction, for two reasons. First, the pseudoplural items were heterogeneous. Some were homophonous with true plurals of other nouns (*cruise*/*crews*, *bruise*/*brews*), while others resembled a plural but differed from it in segmental content (*kiss*/*kits*, *toss*/*talks*). Items of the latter type do not support a plural-marked controller-compatible parse of the string; their failure to attract demonstrates only that a plural-like final sibilant is not sufficient to trigger interference. Second, even for the homophonous items, the controller-compatible parse differs in a crucial respect from the reanalyses available in the Russian and Czech cases. A Russian genitive singular such as *večerinki*, reparsed as a nominative plural, remains the same lexical item, 'party'; the two parses differ only in the morphosyntactic value of the ending. A parse of *cruise* as *crew*+/z/, in contrast, entails the substitution of a distinct lexical item with unrelated semantics. Any activation that the controller-compatible parse receives from the surface form may therefore be counteracted by the semantic mismatch between the two lexemes. The absence of pseudoplural attraction is thus consistent with the controllerhood account, but it does not constitute a minimal test of that account. Such a test requires form overlap that preserves lexical identity, a configuration that English number morphology does not supply. The Turkish suffix *-lAr* provides precisely this configuration: *gördükleri* retains the stem *gör-* and its meaning under both the agreement parse and the plural parse, so that the observed null isolates the contribution of controller association without a confounding lexical mismatch.

Previous English studies, however, offer two more studies that bear on the controllerhood account more directly. One such study was conducted by Nicol et al. (2016). In their production experiment, they manipulated plurality on possessors, as illustrated in (16). Plurality on the object noun *gardens* induced attraction when (16b) is compared with (16a), and when (16d) is compared with (16c). However, plurality on the possessor *elves'* did not induce additional attraction: the comparison of (16c) with (16a), and of (16d) with (16b), yielded no reliable increase. This is not a direct comparison with our materials, but the logic is parallel. English possessive-marked nominals can occur inside subject position, yet they never control verbal agreement in English; possessive morphology therefore cannot become statistically associated with controllerhood. As Lago et al. (2019) argued, this constitutes the critical difference between the two languages: Turkish genitives regularly control agreement on embedded verbs and induce attraction, whereas English possessives never control agreement and do not.

(16) a. The statue in the elf's garden ...

b. The statue in the elf's gardens ...

c. The statue in the elves' garden ...

d. The statue in the elves' gardens ...

Recent comprehension evidence from English narrows the space of what surface form can contribute. Brehm et al. (2020) manipulated the morpho-orthography of attractors in self-paced reading, comparing typically marked nouns with atypical ones on both sides of the number divide: singular nouns whose form resembles a plural (*dress*, *cactus*) and plural nouns lacking the *-s* ending (*men*, *cacti*). Morpho-orthography affected processing at the verb, i.e. atypical plurals slowed reading regardless of grammaticality, but it did not modulate attraction: plural-looking singulars did not induce attraction, and plurals without plural-typical orthography attracted nonetheless. Brehm and colleagues conclude that what is retrieved at the verb is the noun's number feature, not its form—in their words, "features of words, not words themselves, are retrieved." Our Turkish results support the same claim, arguing against direct phonological modulation. The surface form can burden lexical and verb processing, but the representation relevant for agreement is the abstract number specification.

5.3 Interpreting Null Effects as a Primary Result

The core theoretical claim of this paper rests on a null effect—the absence of attraction from verbal attractors—and so it is important to discuss the evidential standards applied here. We treated the absence of verbal attraction as a target estimand rather than a by-product of non-significance, and three lines of evidence support this treatment.

First, the null replicates across experiments with different condition structures. In Experiment 1, verbal attractors were interleaved with nominal attractors that produced robust attraction; in Experiment 2, only verbal conditions were retained. The verbal null was the same in both designs.

Replication across designs rules out explanations tied to one experimental context, such as the possibility that exposure to strong nominal attraction in Experiment 1 masked a weak verbal effect. Experiment 2 removes nominal items entirely, and the null persists. Experiment 3 extends the replication across measures: the same verbal manipulation produced no attraction signature in word-by-word reading times.

Second, the null is coherent across descriptive and model-based analyses. In all three experiments, ungrammatical verbal conditions show very similar acceptance rates across attractor number, and the model-implied estimates for verbal attraction are centered near zero. Furthermore, the Experiment 1 follow-up analysis restricted to verbal trials directly addresses the most relevant question: whether verbal attraction is present at all, not merely whether it is smaller than nominal attraction. That analysis, too, found no evidence for attraction.

Third, the samples are large by the standards of the attraction literature, and the null is quantified with formal model comparison. After exclusions, Experiments 1–3 included 92, 78, and 73 participants, respectively—sample sizes that are substantially larger than those in prior Turkish attraction studies (Lago et al., 2019; Türk and Logačev, 2024) and comparable to or exceeding those in most published attraction experiments cross-linguistically. To move beyond a simple absence of significance, we computed Bayes Factors via bridge sampling for the verbal-only follow-up model in Experiment 1, comparing a model that included the critical Grammaticality × Attractor Number interaction to one that omitted it. The resulting BF_{01} = 4.73 indicated moderate evidence in favor of the null model: the data were approximately five times more likely under a model without verbal attraction than under one with it. Because Bayes Factors are sensitive to the prior distribution, we conducted a prior-sensitivity analysis in which we refit the same model pair under five increasingly concentrated zero-centered normal priors on the fixed effects ($\mathcal{N}(0, 2.0)$ through $\mathcal{N}(0, 0.1)$). The null was favored across all five priors (BF_{01} ranging from 2.73-12.40), confirming that the conclusion does not depend on a particular prior specification.

We computed the same quantity for Experiment 3, where all three critical comparisons favored the null: the acceptability interaction (BF_{01} = 3.11), the reading-time interaction at the critical verb

(BF_{01} = 36.27), and the reading-time interaction at the spillover region (BF_{01} = 12.26). These comparisons place equal prior mass on both signs of the interaction, but attraction predicts a particular sign: on the acceptability scale it predicts a negative interaction, since attraction raises acceptance of ungrammatical sentences with plural attractors, whereas on the reading-time scale it predicts a positive one, since attraction speeds those sentences up. Restricting the alternative to the predicted direction, so that it is not penalized for prior mass on a direction the account excludes, leaves the conclusion unchanged and strengthens it at the two reading-time regions (BF_{01} = 3.67, 26.78, and 91.44, respectively). The spillover region gives the strongest evidence of the three precisely because its posterior mass lies almost entirely on the side opposite to attraction.

Finally, we note that the null does not imply that the true effect is mathematically zero. It implies that any residual effect, if present, is not strong enough to support the theoretical prediction that surface-form overlap alone should generate attraction. A phonological modulation account now needs to specify conditions under which verbal overlap does produce attraction. Our design was constructed to provide such conditions, and no effect emerged.

Two additional methodological concerns deserve comment. One is whether speeded acceptability judgments might miss transient processing interference that dissipates before the final decision. Experiment 3 was designed to address exactly this concern: if verbal attractors caused fleeting interference during incremental parsing, it should have surfaced as reading-time modulation at the critical verb or the spillover region. We found no facilitatory modulation at either region; the only attractor-related posterior mass was a weak, non-credible slowdown in the spillover region, opposite in direction to attraction. Any residual transient effect would therefore have to be weak enough to escape both word-by-word reading times and final acceptability, although measures with finer temporal resolution, such as eye-tracking or EEG, could in principle still detect it. A second concern is whether participants might simply have ignored verbal morphology altogether—a shallow-processing account. This is unlikely for two reasons: participants showed robust sensitivity to verbal agreement in the baseline grammaticality contrasts, reliably rejecting ungrammatical sentences, and Experiment 2 revealed sensitivity to attractor number *within* grammatical conditions, demonstrating that morphological variation on the reduced relative clause was being registered. The null result for verbal attraction therefore cannot be attributed to a general failure to process the attractor's morphology.

Two design limitations should also be acknowledged. First, Experiments 1 and 2 used a 1:1 ratio of experimental to filler items. A higher filler proportion is generally preferable in judgment studies because it better disguises a salient manipulation; Experiment 3 raised the ratio to roughly 1:2, and the pattern of results was unchanged. Second, our reliance on binary speeded acceptability judgments in Experiments 1 and 2 was motivated by comparability with the prior Turkish attraction studies whose effects we aimed to replicate (Lago et al., 2019; Türk and Logačev, 2024), and by the large number of trials the task affords. The convergence of the judgment data with the self-paced reading data in Experiment 3 mitigates concerns tied to any single task.

5.4 Response Bias

One concern with speeded acceptability judgment tasks is that participants differ in their overall willingness to press 'acceptable'—a response bias that is orthogonal to perceptual sensitivity. Hammerly et al. (2019) and Türk (2022) have shown that filler composition and task instructions can shift overall acceptance rates, and that such shifts can obscure the true effect of experimental manipulations in speeded judgment paradigms. Our Experiment 1 data are consistent with such a shift: overall acceptance rates for nominal-attractor sentences were lower than in Türk and Logačev (2024), suggesting that the inclusion of non-attracting verbal trials shifted the response criterion toward a more conservative baseline. Importantly, however, this shift did not substantially reduce the *magnitude* of attraction within the nominal conditions. Exposure to non-attracting verbal trials shifted overall bias without recalibrating the attraction effect itself. Experiment 3 shifted the baseline in the opposite direction: participants accepted ungrammatical sentences at more than twice the rate of Experiments 1 and 2 (mean false alarm rates of 13.2% vs. 3.9% and 5.2%), corresponding to a more liberal mean criterion ($c = -0.03$, against $+0.04$ and $+0.05$ in the verbal conditions of Experiments 1 and 2). If the verbal null in the first two experiments merely reflected a lack of room for false alarms, this more liberal baseline should have given attraction room to appear; it did not.

A remaining question is whether individual differences in response bias might explain the absence of verbal attraction—that is, whether participants who adopted a more liberal criterion showed verbal attraction effects that were washed out in the group-level analysis. Although we did not design the experiments to test this directly, we conducted a post-hoc analysis relating each participant's response criterion c to their attraction magnitude, defined as the difference in false alarm rates between plural- and singular-attractor ungrammatical conditions. For each participant, we computed the hit rate (accepting grammatical sentences) and false alarm rate (accepting ungrammatical sentences) separately for each attractor type, and from these derived the signal detection criterion $c = -0.5 \times [z(\text{hit rate}) + z(\text{false alarm rate})]$, with rates log-linear corrected as $(k + 0.5)/(n + 1)$ before the z-transformation. We ran this analysis for the verbal conditions of all three experiments and, as benchmarks where attraction is known to be present, for the nominal conditions of Experiment 1 and the nominal-attractor data of Türk and Logačev (2024). Because this analysis is descriptive and post hoc, we report it with simple correlation statistics rather than the model-based framework used elsewhere.

Figure 7 shows subject-level attraction estimates plotted against response criterion for each condition. Where attraction is genuinely present, criterion was reliably correlated with attraction magnitude: participants with a more liberal criterion (lower c) showed larger attraction both for nominal attractors in Experiment 1 ($r = -0.43$, $p < .001$) and in the Türk and Logačev (2024) benchmark data ($r = -0.50$, $p < .001$), consistent with the idea that response bias modulates the observable magnitude of attraction. For verbal attractors, by contrast, no such relationship emerged in any experiment: Experiment 1 ($r = -0.12$, $p = .24$), Experiment 2 ($r = -0.10$, $p = .39$), and Experiment 3 ($r = -0.10$, $p = .38$). The absence of a criterion–attraction relationship in the verbal conditions across all three experiments indicates that the null effect for verbal attraction is not an artifact of conservative responding: even participants who adopted a liberal criterion and accepted ungrammatical sentences at high rates did not show verbal attraction.

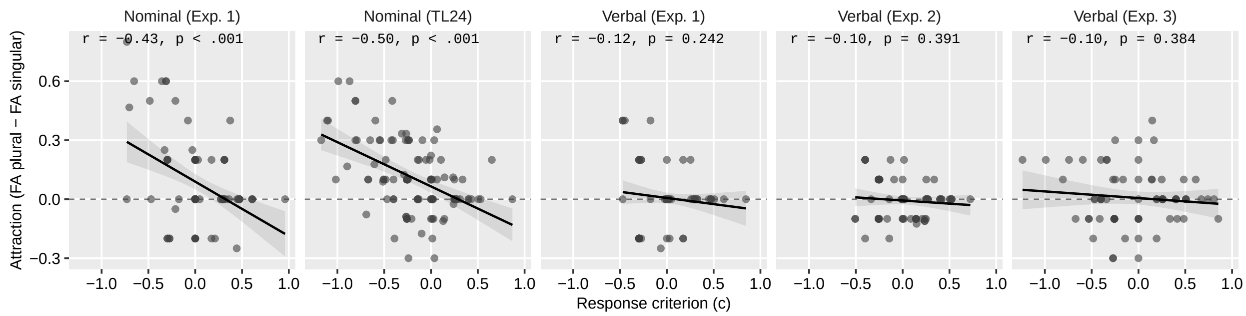


Figure 7: Subject-level attraction estimates as a function of response criterion, shown separately for nominal attractors (Experiment 1 and the nominal data of Türk and Logačev 2024) and verbal attractors (Experiments 1–3). Points represent participants; lines show linear fits with 95% confidence bands.

6 Conclusion

Here we have presented evidence from three experiments in Turkish suggesting that surface-form overlap alone is insufficient to drive agreement illusions. Our central finding is a dissociation between attractors that share the same plural surface form: nominal attractors produce attraction, while verbal attractors do not—neither in sentence-final acceptability judgments nor in word-by-word reading times. This null result is theoretically informative because the design maximized the conditions under which phonological modulation should have produced an effect: the attractor is categorially nominal, the *-lAr* suffix is string-ambiguous for number, and phonological overlap in memory is well-established as a source of interference. The same conclusion is independently supported by the English pseudoplural and Czech null results, while the apparent Russian exception can be explained by controller association rather than by phonology. Agreement attraction in comprehension is therefore better characterized as grammar-constrained memory retrieval than as form-driven confusion. The parser is sensitive to fine-grained distinctions in the morphosyntactic status of potential competitors that are invisible on the surface of the string.

CRediT authorship contribution statement

Utku Türk: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

A Results from additional models for Experiment 1

This appendix reports the model formulas and full fixed-effect posteriors for the three acceptability models of Experiment 1 referenced in the main text: the model restricted to verbal (RC) trials, the $2 \times 2 \times 2$ model over the Experiment 1 data, and the combined model that additionally includes the nominal-attractor data of Türk and Logačev (2024). All three are Bayesian hierarchical logistic regressions (Bernoulli family, logit link) fit in `brms`, with the binary 'acceptable' response as the dependent variable, sum-coded (± 0.5) two-level predictors, and maximal random-effects structures. Priors were $\mathcal{N}(0.85, 0.70)$ on the intercept, $\mathcal{N}(0, 1)$ on fixed-effect slopes, Exponential(1) on random-effect standard deviations, and LKJ(2) on random-effect correlations. Each model was estimated with four chains of 12,000 iterations (2,000 warmup). In all tables, G is Grammaticality (Grammatical = +0.5, Ungrammatical = −0.5) and A is Attractor Number (Plural = +0.5, Singular = −0.5); a two-level slope is thus the full difference between the two levels. Whisker plots show the posterior mean and 95% CrI; the scale of each table is chosen to cover its own estimates, is stated in its caption, and is labeled by the tick marks. The dashed line marks zero.

A.1 Verbal-only model

The follow-up model restricted to the verbal-attractor trials (1,816 observations from the 92 analyzed participants) had the formula

```
yes ~ G * A + (1 + G * A | subject) + (1 + G * A | item)
```

Table 7 reports its fixed effects.

Table 7: Posterior summaries of the fixed effects in the verbal-only model for Experiment 1. Whiskers are drawn on a scale from −2 to 8.6.

Effect	β	95% CrI	$P(\beta > 0)$
Grammaticality (G): Gram vs. Ungram	7.13	[6.31 , 8.09]	> 0.99
Attractor Number (A): Pl vs. Sg	−0.16	[−0.79 , 0.43]	0.31
G × A	−0.18	[−1.36 , 1.02]	0.37

A.2 Full $2 \times 2 \times 2$ model

The model over all Experiment 1 trials (3,623 observations) had the formula

```
yes ~ G * A * T + (1 + G * A * T | subject) + (1 + G * A * T | item)
```

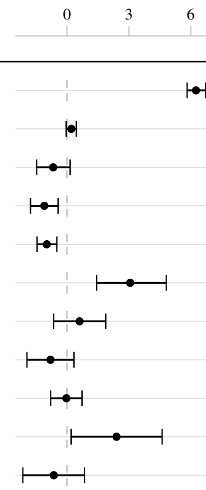
where T is Attractor Type, coded Verbal = +0.5, Nominal = −0.5. The full fixed-effect posteriors of this model are reported in the main text (Table 4) and are not repeated here.

A.3 Combined model with the TL24 data

The combined model added the nominal-attractor data of Türk and Logačev (2024) (7,848 observations in total, from 200 participants and 80 items) and had the same formula as the $2 \times 2 \times 2$ model, except that Attractor Type now had three levels coded by two mean-centered contrasts, as described in the main text: T1 contrasts the verbal condition against the mean of the two nominal conditions (v = 0.333, nom.cur = nom.tl24 = −0.167), and T2 contrasts the two nominal conditions with each other (nom.cur = 0.333, nom.tl24 = −0.333, v = 0). Positive T1 coefficients thus indicate higher values in the verbal condition. Table 8 reports its fixed effects; the derived per-condition attraction contrasts shown in Figure 2 were computed from these posteriors as described in the main text.

Table 8: Posterior summaries of the fixed effects in the combined Experiment 1 model including the Türk and Logačev (2024) data. Whiskers are drawn on a scale from −2.5 to 7.

Effect	β	95% CrI	$P(\beta > 0)$
Grammaticality (G): Gram vs. Ungram	6.25	[5.82 , 6.72]	> 0.99
Attractor Number (A): Pl vs. Sg	0.21	[−0.04 , 0.45]	0.95
T1: Verbal vs. Nominal (pooled)	−0.67	[−1.47 , 0.15]	0.05
T2: Nominal-Cur. vs. Nominal-TL24	−1.10	[−1.77 , −0.43]	< 0.01
G × A	−0.98	[−1.45 , −0.49]	< 0.01
G × T1	3.06	[1.44 , 4.81]	> 0.99
G × T2	0.61	[−0.65 , 1.88]	0.83
A × T1	−0.80	[−1.94 , 0.34]	0.08
A × T2	−0.03	[−0.79 , 0.73]	0.46
G × A × T1	2.40	[0.20 , 4.61]	0.98
G × A × T2	−0.64	[−2.14 , 0.85]	0.20



B Experiment 3: reading-time descriptives and per-region model estimates

This appendix reports, for each of the eight sentence regions of Experiment 3, the descriptive reading times by condition (Table 9) and the full fixed-effect posteriors of the per-region reading-time models (Table 10). Each region was fit with a Bayesian hierarchical model of log reading time (Gaussian family) with the formula

```
log_rt ~ G * A + (1 + G * A | subject) + (1 + G * A | item)
```

with G and A sum-coded as in the acceptability models (Grammatical = +0.5, Ungrammatical = −0.5; Plural = +0.5, Singular = −0.5), so a slope is the full difference between the two levels. Priors were $\mathcal{N}$(region mean log RT, 0.5) on the intercept, $\mathcal{N}(0, 1)$ on slopes, half-normal $\mathcal{N}(0, 1)$ on standard deviations and the residual scale, and LKJ(2) on random-effect correlations; each model was estimated with four chains of 12,000 iterations (2,000 warmup).

Table 9: Mean reading times in milliseconds (with standard errors) by region and condition in Experiment 3. Gram = grammatical, Ung = ungrammatical; PL and SG denote plural and singular attractors. $n = 740$ trials per condition cell; SEs are between-trial standard deviations divided by $\sqrt{n}$.

		Grammatical		Ungrammatical	
Pos	Region	PL	SG	PL	SG
1	Adjunct	595 (48)	548 (23)	537 (25)	634 (103)
2	Attractor	676 (20)	582 (13)	623 (15)	556 (12)
3	Head	582 (12)	578 (14)	554 (12)	563 (18)
4	Adjunct	539 (12)	543 (11)	521 (10)	542 (11)
5	Adjunct	496 (12)	507 (12)	500 (11)	494 (10)
6	Critical Verb	499 (11)	510 (12)	547 (12)	563 (13)
7	Spillover	456 (11)	452 (11)	509 (23)	446 (8)
8	Matrix Verb	810 (32)	831 (35)	802 (35)	861 (39)

Table 10: Fixed-effect posteriors of the per-region log reading-time models in Experiment 3: posterior mean (β), 95% CrI, and the posterior probability that the coefficient is positive (P). Coefficients are on the log-ms scale. Intercepts are omitted; they correspond to the grand mean log reading time of each region.

		Grammaticality (G)			Attractor Number (A)			G × A		
Pos	Region	β	95% CrI	P	β	95% CrI	P	β	95% CrI	P
1	Adjunct	0.013	[−0.020, 0.047]	.79	−0.009	[−0.039, 0.020]	.27	0.013	[−0.063, 0.089]	.64
2	Attractor	0.038	[0.006, 0.071]	.99	0.090	[0.055, 0.125]	>.99	−0.005	[−0.067, 0.057]	.44
3	Head	0.036	[0.005, 0.066]	.99	0.012	[−0.018, 0.042]	.79	0.004	[−0.052, 0.059]	.56
4	Adjunct	0.014	[−0.012, 0.039]	.86	−0.021	[−0.047, 0.004]	.05	0.026	[−0.026, 0.078]	.84
5	Adjunct	−0.000	[−0.026, 0.026]	.50	−0.009	[−0.033, 0.014]	.22	−0.019	[−0.075, 0.036]	.24
6	Critical Verb	−0.083	[−0.113, −0.053]	<.01	−0.017	[−0.047, 0.014]	.13	0.013	[−0.041, 0.067]	.68
7	Spillover	−0.023	[−0.050, 0.006]	.06	0.028	[0.003, 0.053]	.98	−0.043	[−0.099, 0.013]	.07
8	Matrix Verb	0.027	[−0.039, 0.093]	.79	−0.020	[−0.071, 0.030]	.21	0.025	[−0.086, 0.135]	.68

Data Availability

Materials, code and data for following experiments available at: https://osf.io/p6243/overview?view_only=a4b4f0e64d924d48b68782169d646381

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